

Cisco CCNA Laboratoriya Layihəsi

Müəssisə Şəbəkəsinin Qurulması və Konfiqurasiyası

Hazırlayan: Hüseyn Asirov

İxtisas: Şəbəkə Mühəndisliyi / Kibertəhlükəsizlik

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- 21. Access Control List (ACL)**
- 22. Nəticə**

Layihə haqqında:

Bu layihə Cisco CCNA proqramı çərçivəsində öyrənilən əsas şəbəkə texnologiyalarının praktik tətbiqini əhatə edir. Layihənin məqsədi müəssisə şəbəkəsinin qurulması, idarə olunması, təhlükəsizliyinin təmin edilməsi və yaranan

problemlərin aradan qaldırılması üzrə praktiki bacarıqların nümayiş etdirilməsidir.

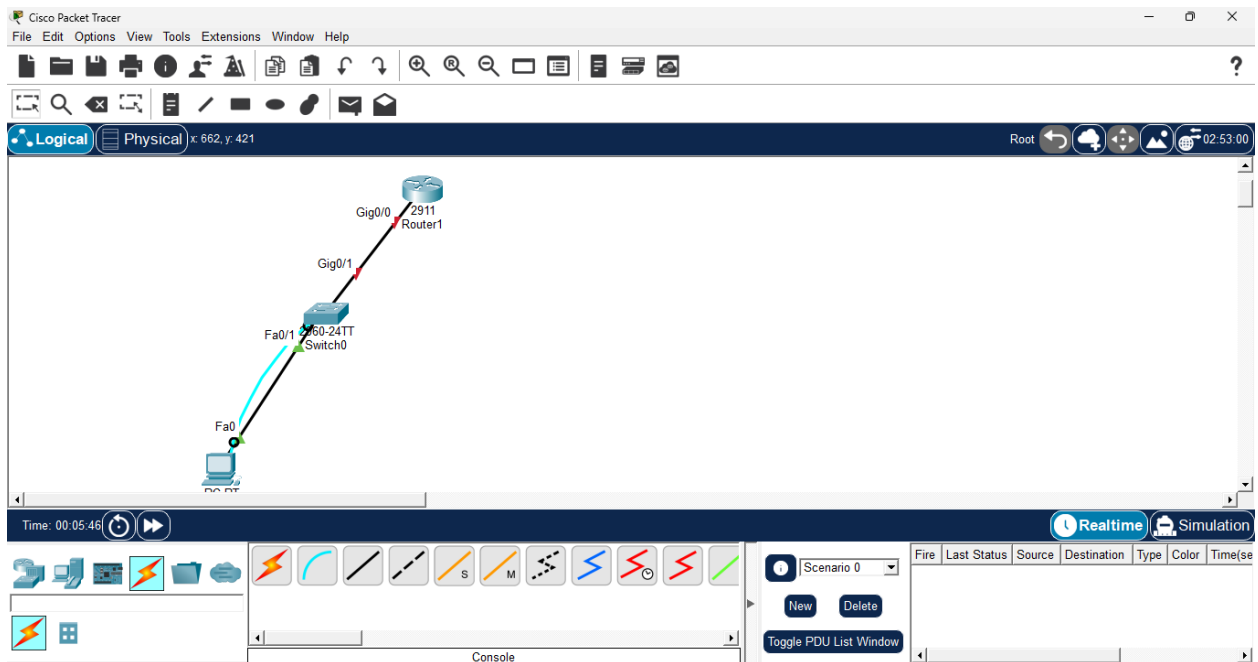
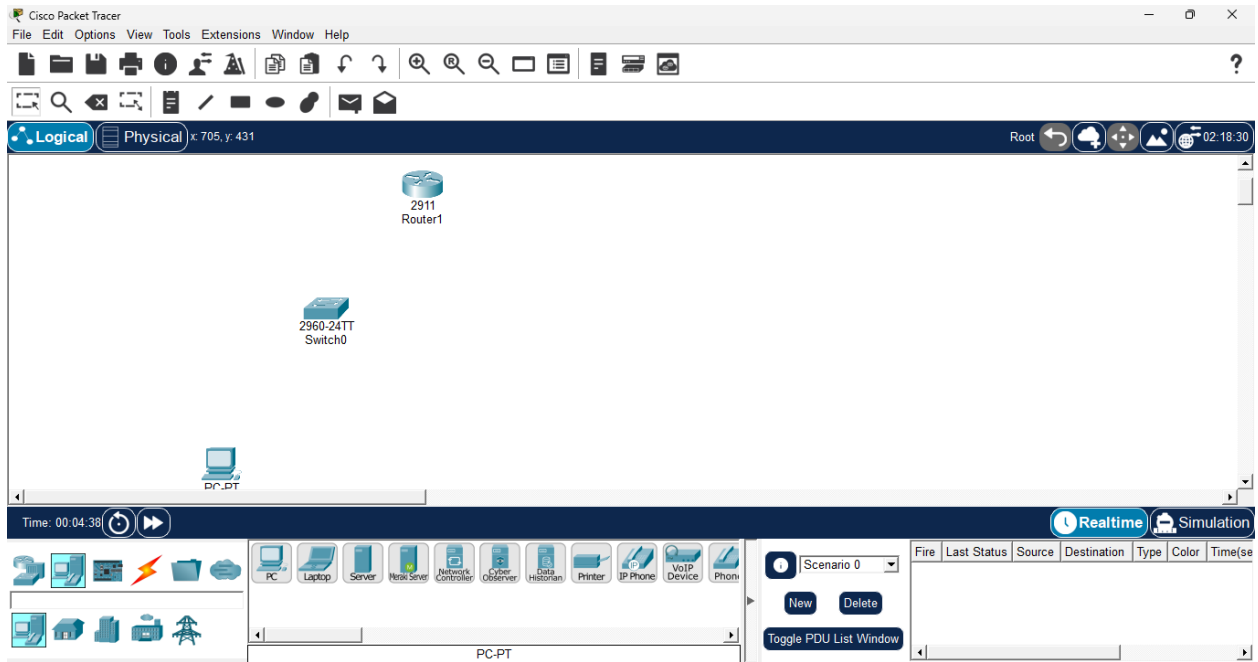
Layihə boyunca Cisco IOS əmrlərindən istifadə edilərək router və switch cihazlarının konfigurasiyası həyata keçirilmiş, şəbəkənin təhlükəsiz və stabil işləməsi üçün müxtəlif texnologiyalar tətbiq olunmuşdur.

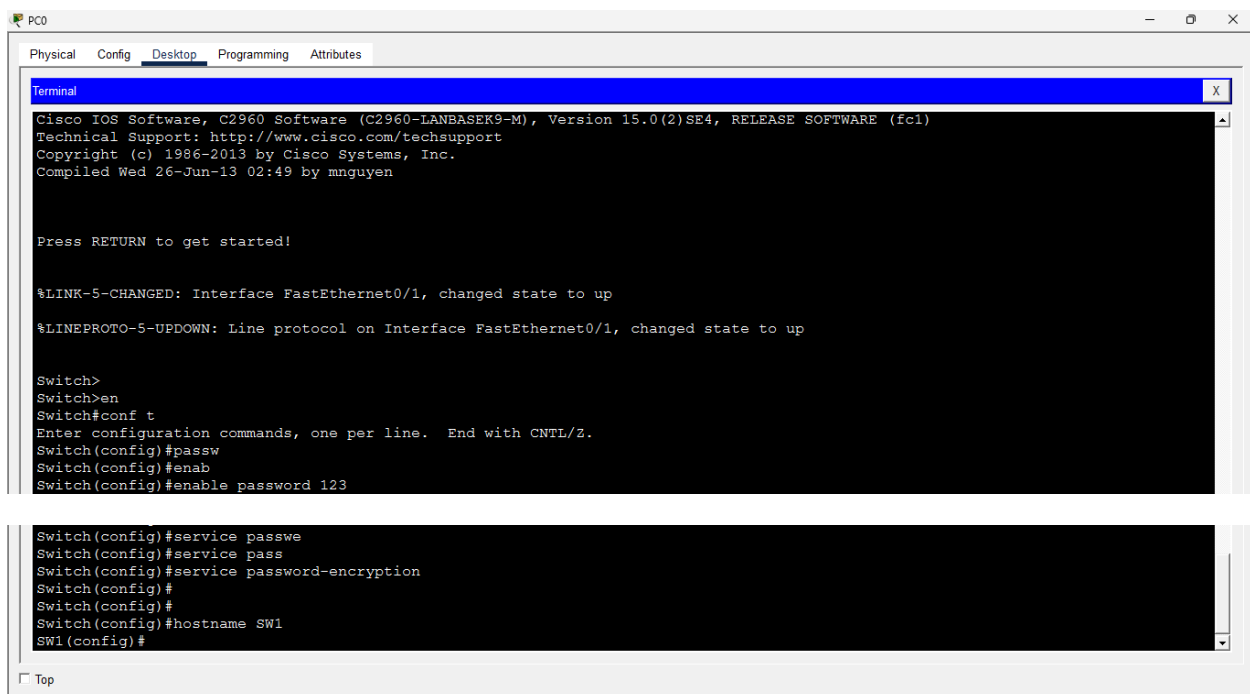
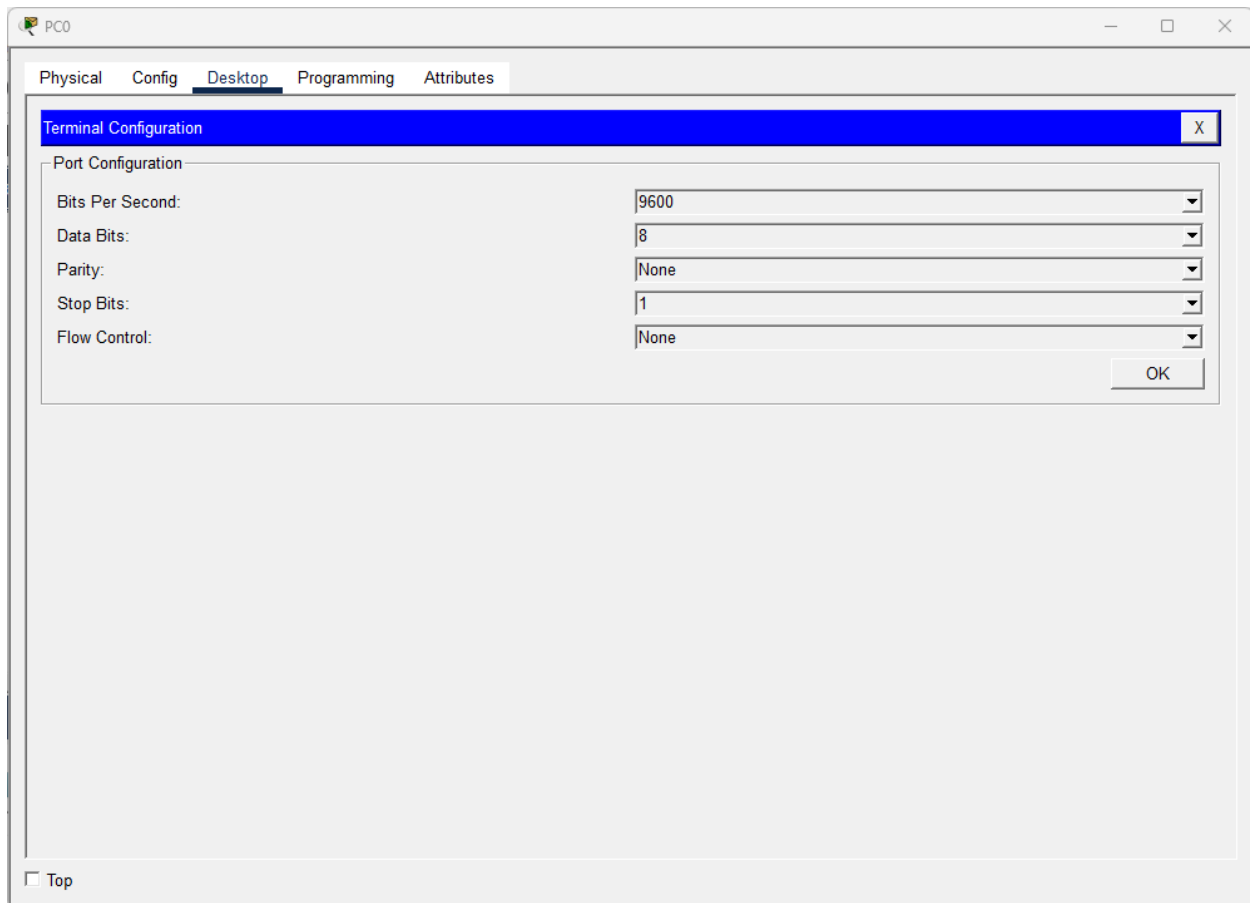
İstifadə olunan texnologiyalar

- Cisco Packet Tracer**
- Cisco IOS**
- Router**
- Switch**
- VLAN**
- Inter-VLAN Routing**
- EtherChannel**
- Trunk**
- VTP**

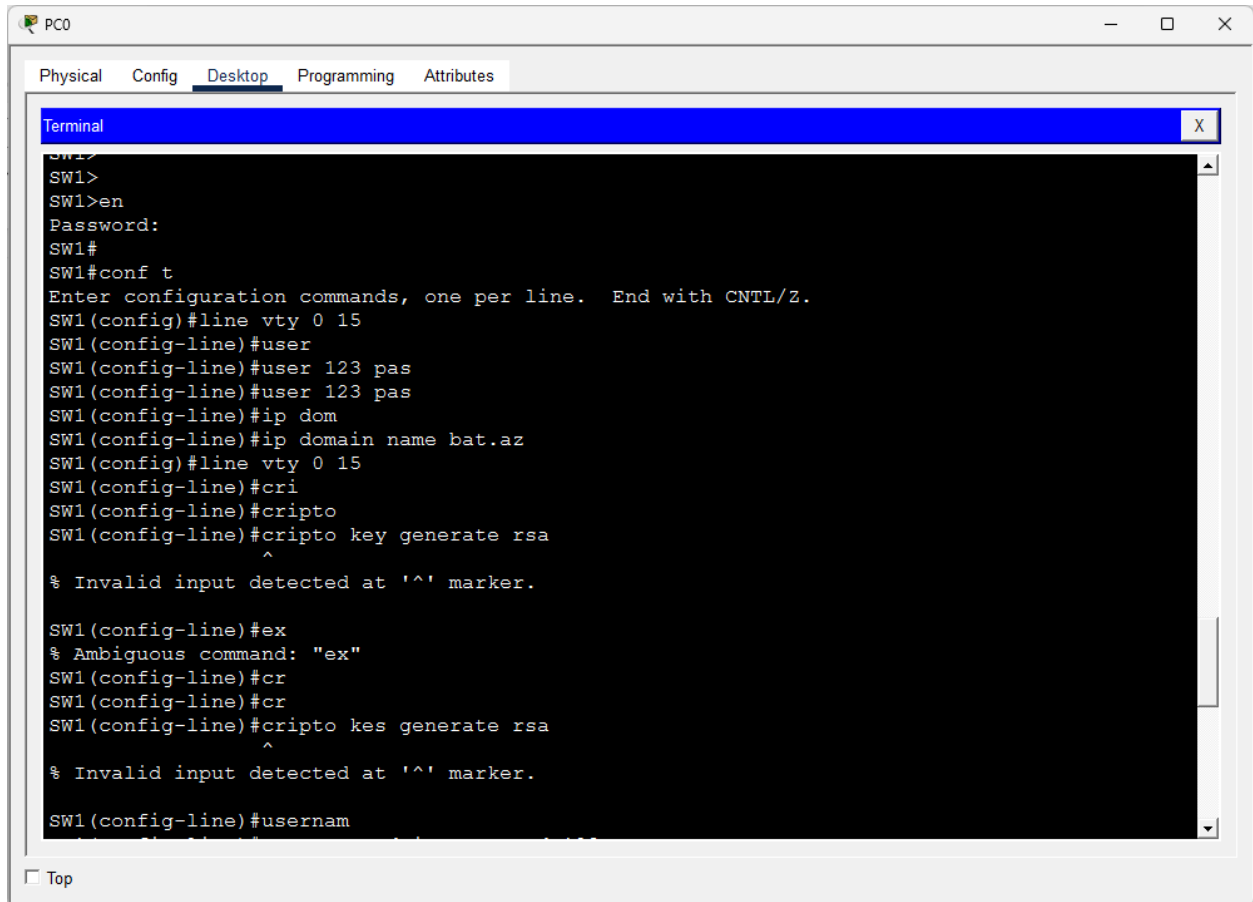
- **DHCP**
- **EIGRP**
- **Spanning Tree Protocol (STP)**
- **Port Security**
- **SSH**
- **Access Control List (ACL)**

Switch initial konfigurasiya ucun arxitekturanin qurulmasi





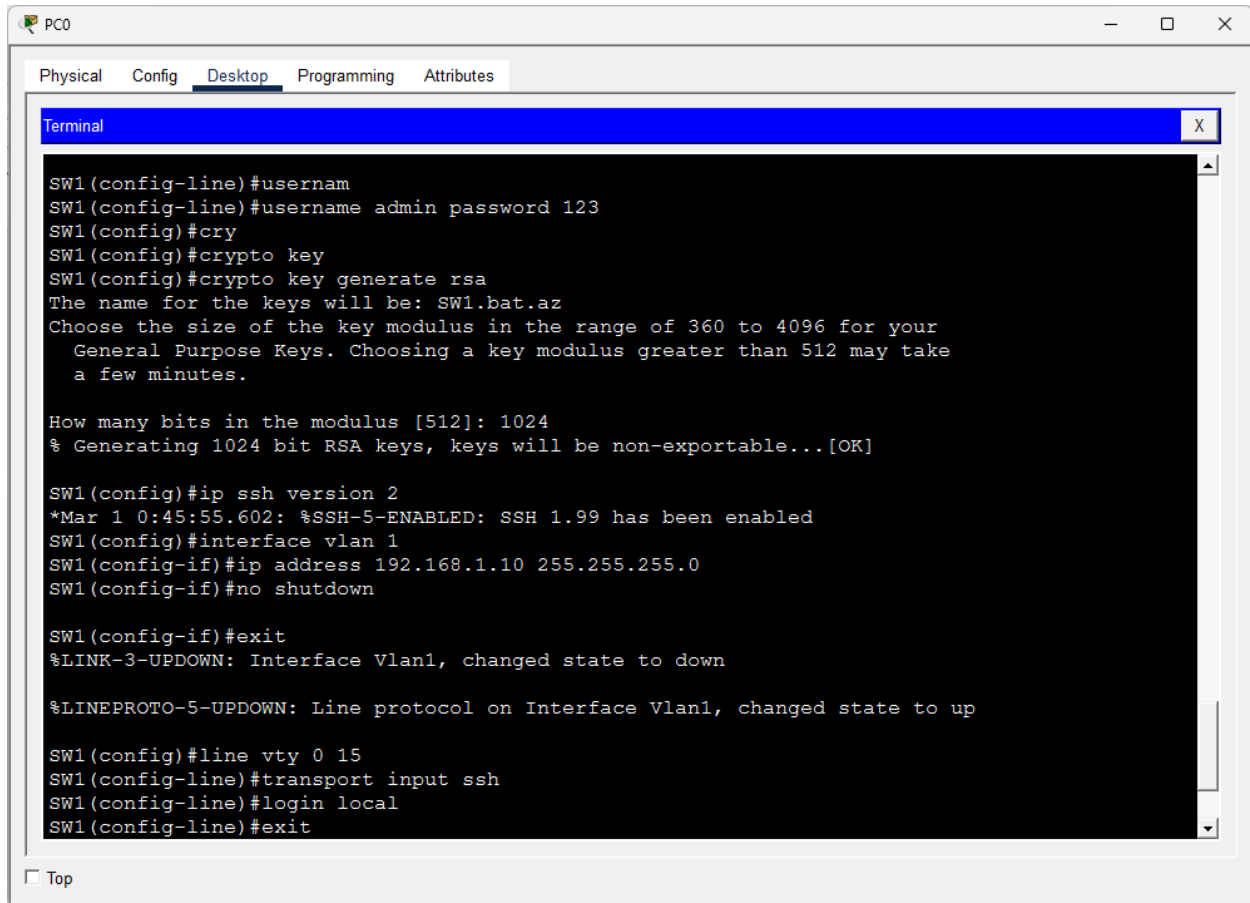
Switchde ssh servisinin qurulmasi telnetin baglanmasi



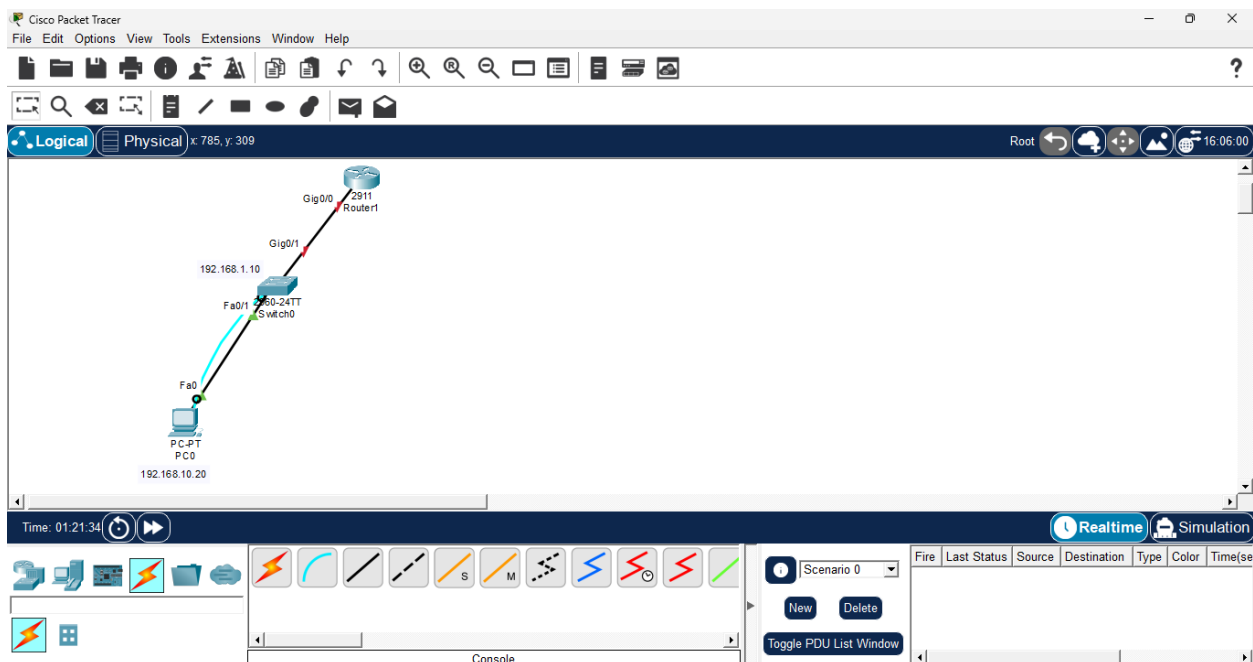
The screenshot shows a PC0 window with a terminal interface. The terminal is titled "Terminal" and has a blue header bar. The user is logged in as "SW1" and is in the "en" (enable) mode. The user enters the command "conf t" to enter configuration mode. The prompt changes to "SW1(config)#". The user enters "line vty 0 15" to configure the VTY lines. The prompt changes to "SW1(config-line)#". The user enters "user 123 pas" to set the username and password. The prompt changes to "SW1(config-line)#". The user enters "ip domain" to set the domain name. The prompt changes to "SW1(config-line)#". The user enters "ip domain name bat.az" to set the domain name. The prompt changes to "SW1(config-line)#". The user enters "line vty 0 15" to configure the VTY lines. The prompt changes to "SW1(config-line)#". The user enters "crypto" to enter the crypto configuration mode. The prompt changes to "SW1(config-line)#". The user enters "crypto key generate rsa" to generate the RSA key. The prompt changes to "SW1(config-line)#". The user enters "ex" to exit the crypto configuration mode. The prompt changes to "SW1(config-line)#". The user enters "cr" to enter the crypto configuration mode. The prompt changes to "SW1(config-line)#". The user enters "crypto kes generate rsa" to generate the RSA key. The prompt changes to "SW1(config-line)#". The user enters "username" to set the username. The prompt changes to "SW1(config-line)#".

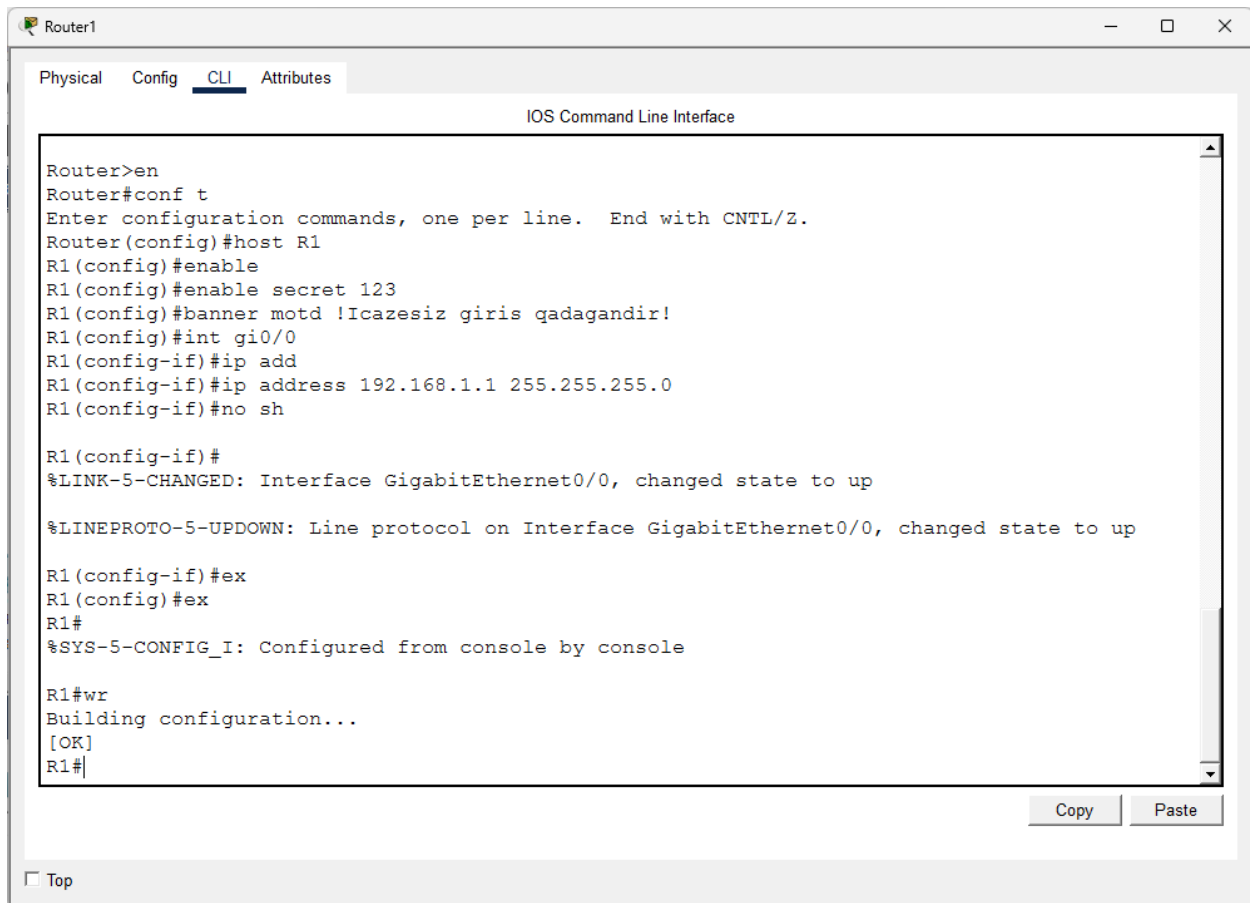
```
SW1>  
SW1>  
SW1>en  
Password:  
SW1#  
SW1#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
SW1(config)#line vty 0 15  
SW1(config-line)#user  
SW1(config-line)#user 123 pas  
SW1(config-line)#user 123 pas  
SW1(config-line)#ip dom  
SW1(config-line)#ip domain name bat.az  
SW1(config)#line vty 0 15  
SW1(config-line)#cri  
SW1(config-line)#cripto  
SW1(config-line)#cripto key generate rsa  
^  
% Invalid input detected at '^' marker.  
  
SW1(config-line)#ex  
% Ambiguous command: "ex"  
SW1(config-line)#cr  
SW1(config-line)#cr  
SW1(config-line)#cripto kes generate rsa  
^  
% Invalid input detected at '^' marker.  
  
SW1(config-line)#usernam
```

☐ Top

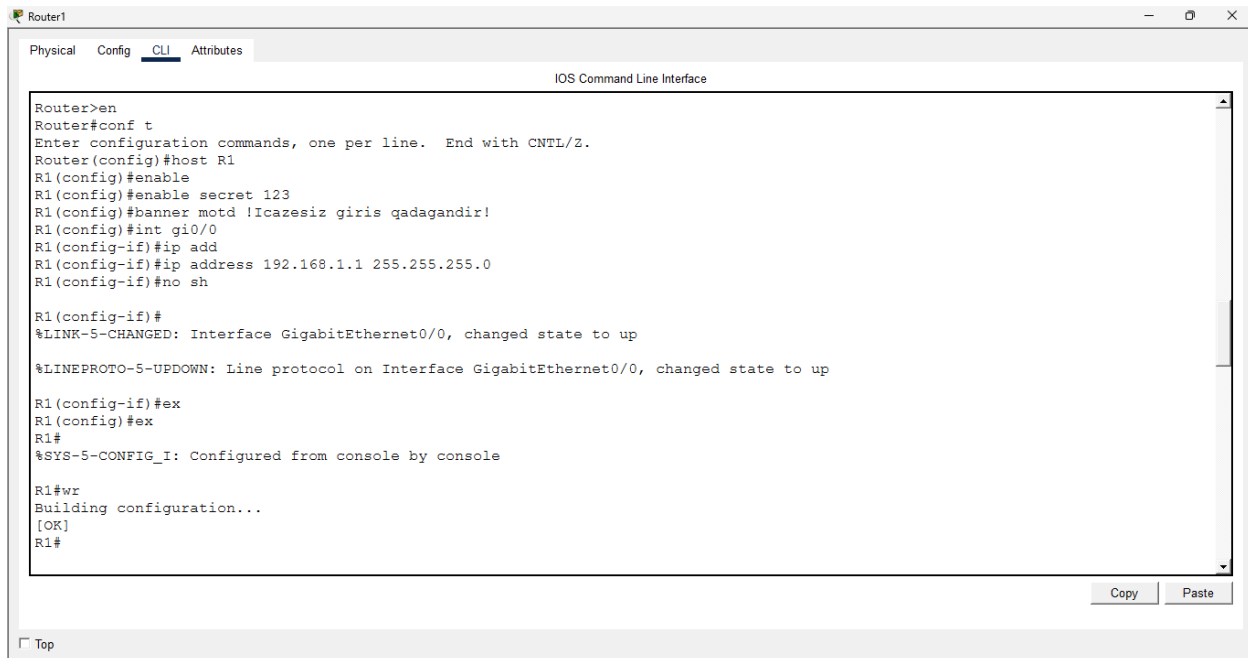


Router initial config ucin arxitekturanin qurulmasi





Routerda ssh servisinin qurulmasi telnetin baglanmasi



The screenshot shows a Cisco Router CLI window titled "Router1". The window has tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is selected, and the title bar indicates "IOS Command Line Interface". The command history shows the following commands and their outputs:

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#host R1
R1(config)#enable
R1(config)#enable secret 123
R1(config)#banner motd !Icazesiz giris qadagandir!
R1(config)#int gi0/0
R1(config-if)#ip add
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no sh

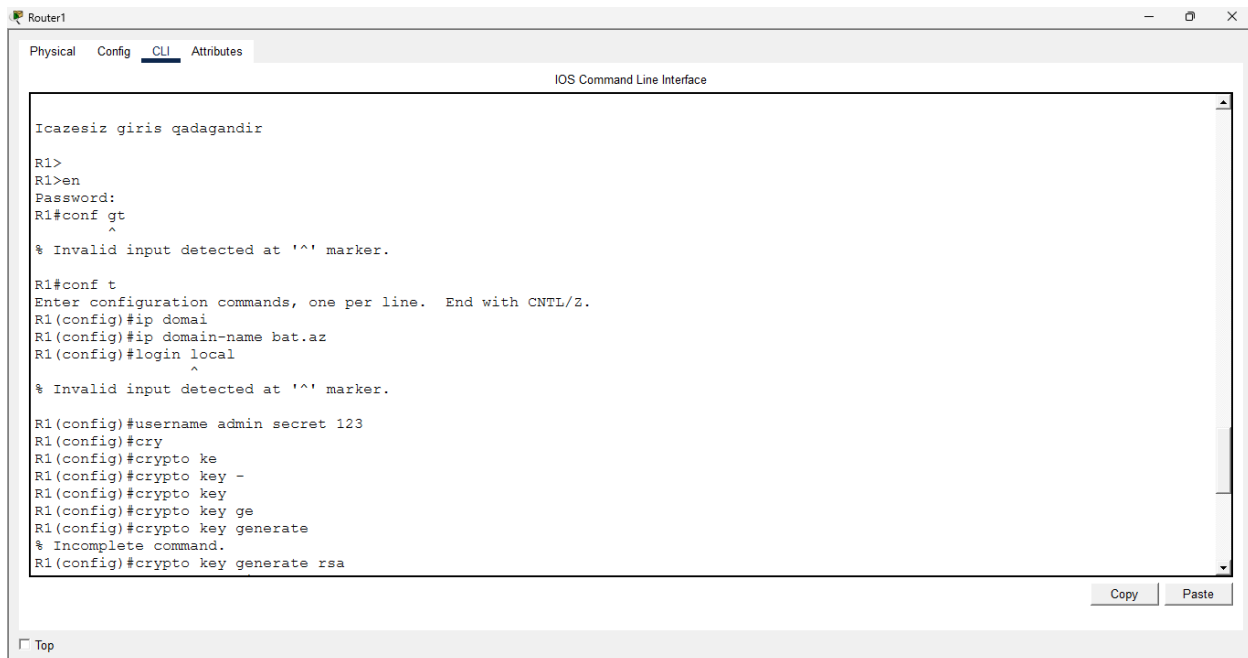
R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#ex
R1(config)#ex
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#wr
Building configuration...
[OK]
R1#
```

At the bottom right of the CLI window, there are "Copy" and "Paste" buttons. At the bottom left, there is a "Top" button.



The screenshot shows a Cisco Router CLI window titled "Router1". The window has tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is selected, and the title bar indicates "IOS Command Line Interface". The command history shows the following commands and their outputs:

```
Icazesiz giris qadagandir

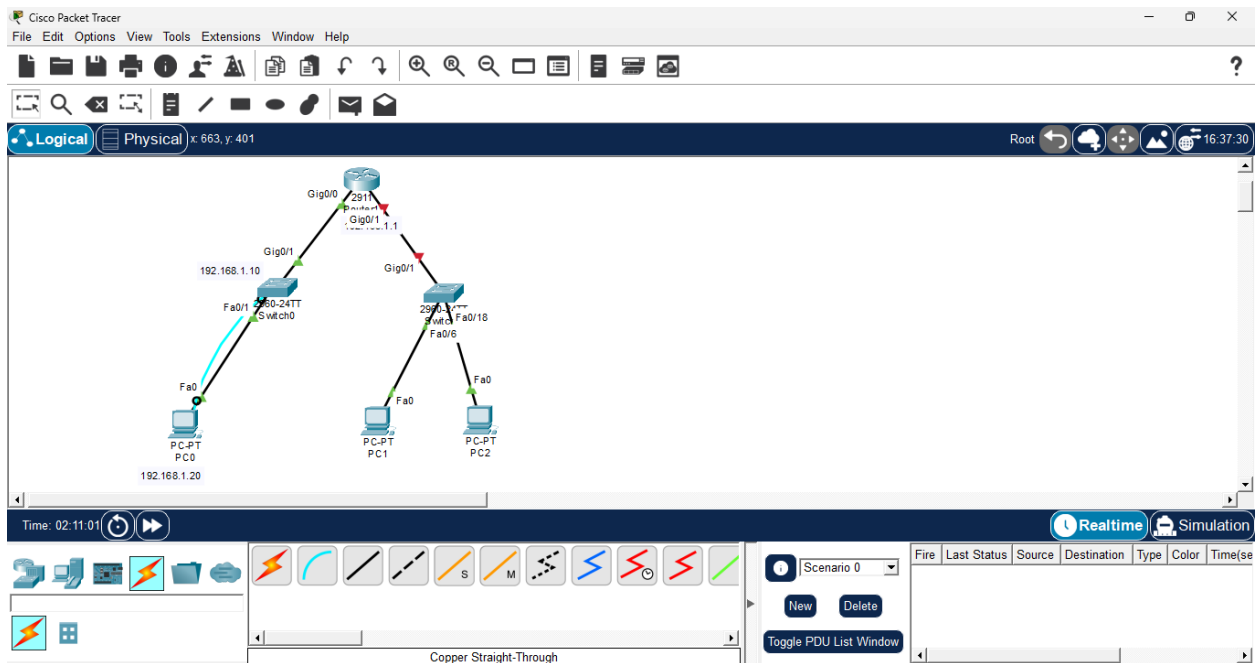
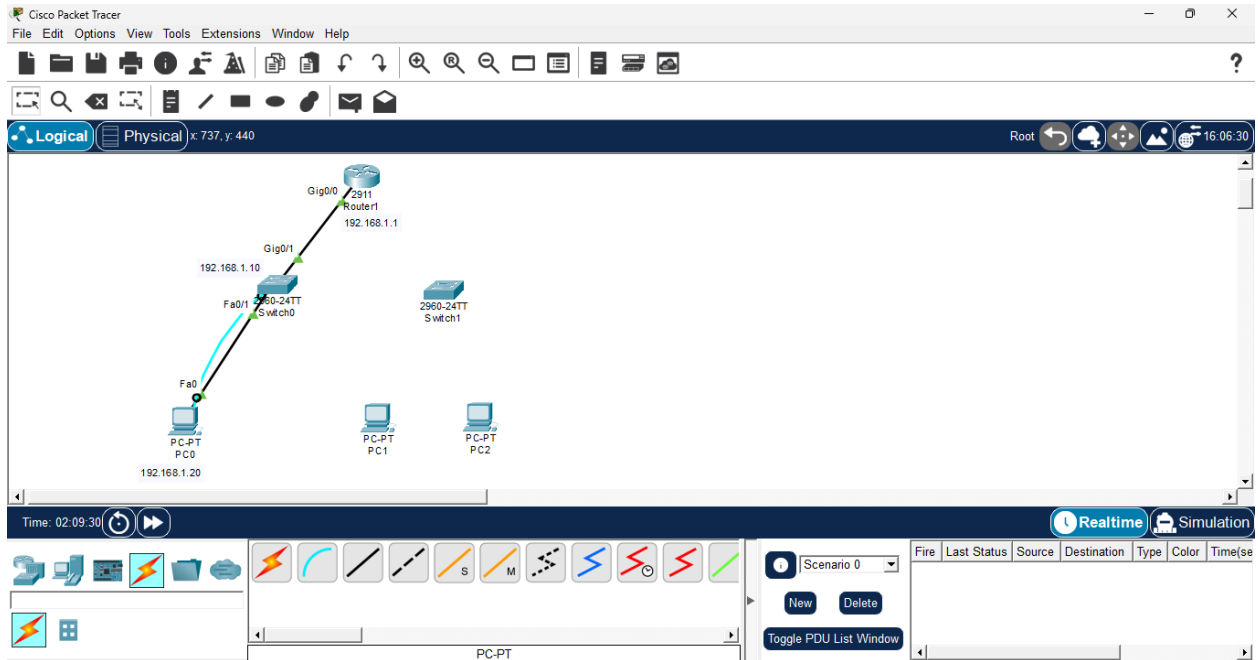
R1>
R1>en
Password:
R1#conf gt
^
% Invalid input detected at '^' marker.

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip domai
R1(config)#ip domain-name bat.az
R1(config)#login local
^
% Invalid input detected at '^' marker.

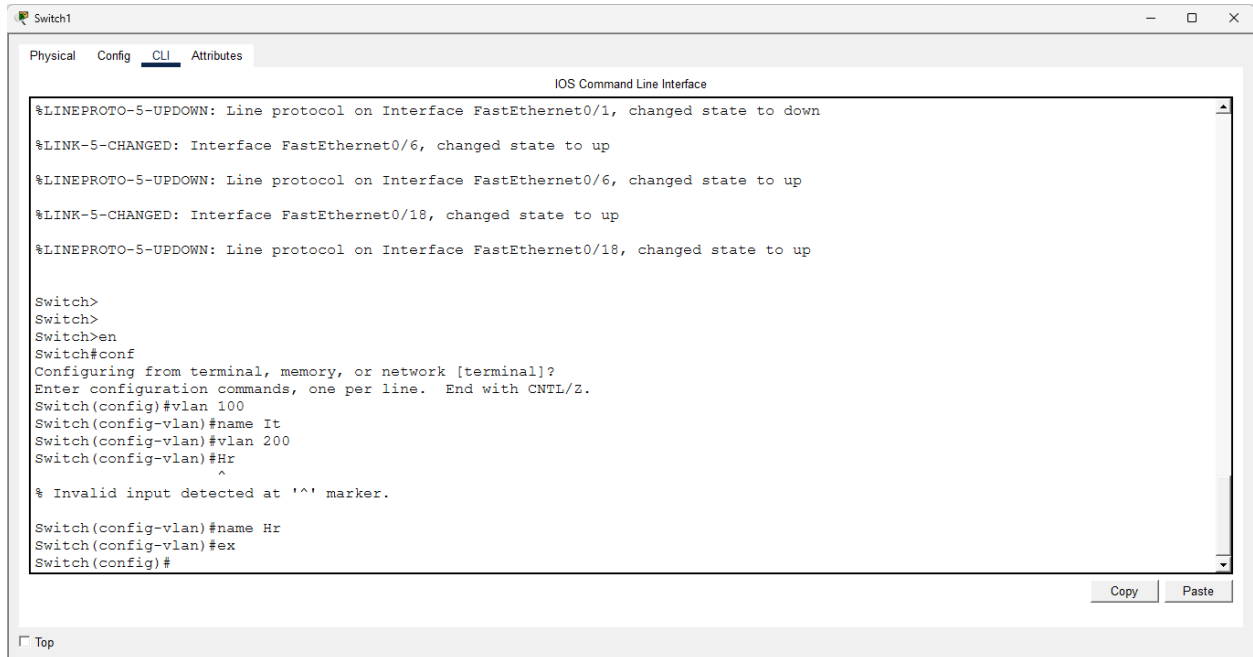
R1(config)#username admin secret 123
R1(config)#cry
R1(config)#crypto ke
R1(config)#crypto key -
R1(config)#crypto key
R1(config)#crypto key ge
R1(config)#crypto key generate
% Incomplete command.
R1(config)#crypto key generate rsa
```

At the bottom right of the CLI window, there are "Copy" and "Paste" buttons. At the bottom left, there is a "Top" button.

Vlanlar ucun yeni arxitekturanin qurulmasi



Vlanların konfigürasyonu



Switch1

Physical Config CLI Attributes

IOS Command Line Interface

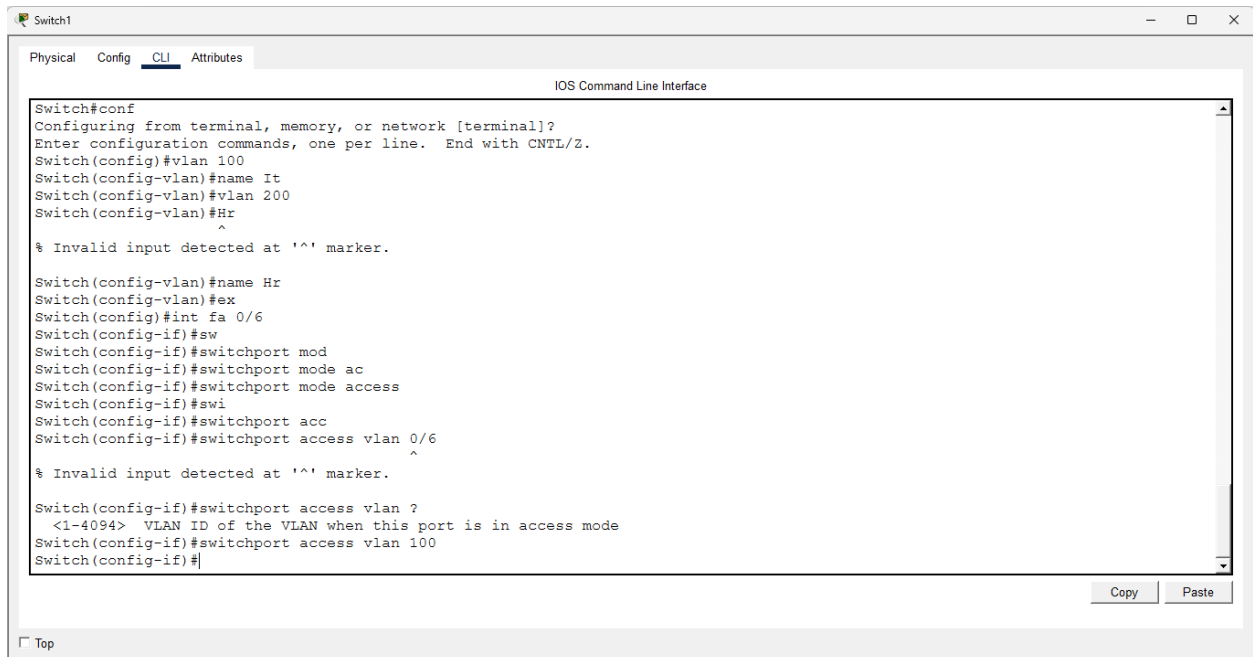
```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/18, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/18, changed state to up

Switch>
Switch>
Switch>en
Switch#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 100
Switch(config-vlan)#name It
Switch(config-vlan)#vlan 200
Switch(config-vlan)#Hr
^
% Invalid input detected at '^' marker.

Switch(config-vlan)#name Hr
Switch(config-vlan)#ex
Switch(config)#
```

Copy Paste

Top



Switch1

Physical Config CLI Attributes

IOS Command Line Interface

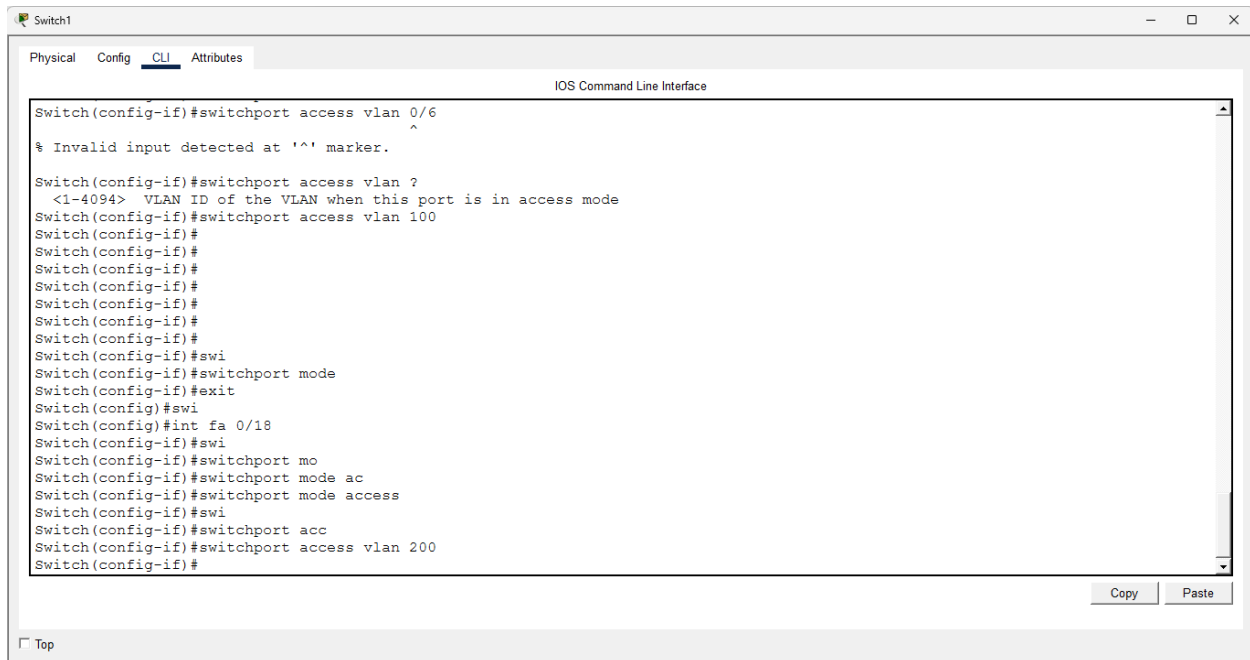
```
Switch#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 100
Switch(config-vlan)#name It
Switch(config-vlan)#vlan 200
Switch(config-vlan)#Hr
^
% Invalid input detected at '^' marker.

Switch(config-vlan)#name Hr
Switch(config-vlan)#ex
Switch(config)#int fa 0/6
Switch(config-if)#sw
Switch(config-if)#switchport mod
Switch(config-if)#switchport mode ac
Switch(config-if)#switchport mode access
Switch(config-if)#swi
Switch(config-if)#switchport acc
Switch(config-if)#switchport access vlan 0/6
^
% Invalid input detected at '^' marker.

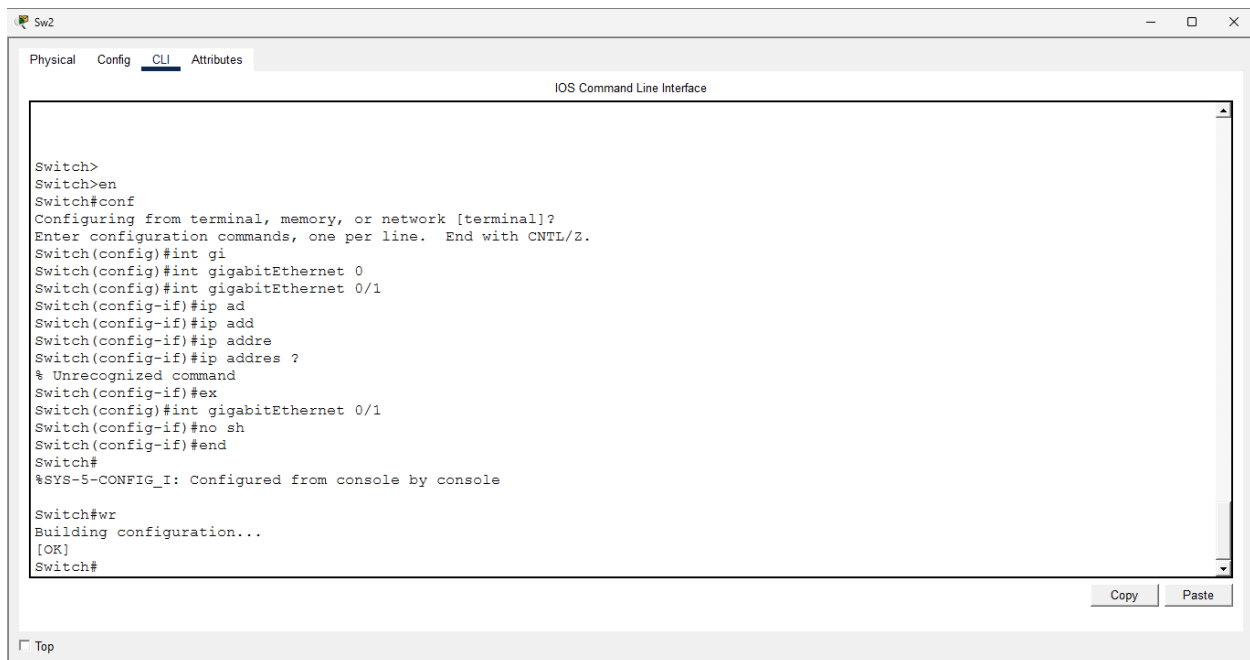
Switch(config-if)#switchport access vlan ?
<1-4094> VLAN ID of the VLAN when this port is in access mode
Switch(config-if)#switchport access vlan 100
Switch(config-if)#
```

Copy Paste

Top



Intervlan Routing



Sw2

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch(config-if)#ip ad
Switch(config-if)#ip add
Switch(config-if)#ip addre
Switch(config-if)#ip addres ?
% Unrecognized command
Switch(config-if)#ex
Switch(config)#int gigabitEthernet 0/1
Switch(config-if)#no sh
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr
Building configuration...
[OK]
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int gi 0/1
Switch(config-if)#swi
Switch(config-if)#switchport mod
Switch(config-if)#switchport mode tr
Switch(config-if)#switchport mode trunk
Switch(config-if)#sw
Switch(config-if)#switchport tr
Switch(config-if)#switchport trunk al
Switch(config-if)#switchport trunk allowed vlan 100,200
Switch(config-if)#
```

Copy Paste

Top

Router1

Physical Config CLI Attributes

IOS Command Line Interface

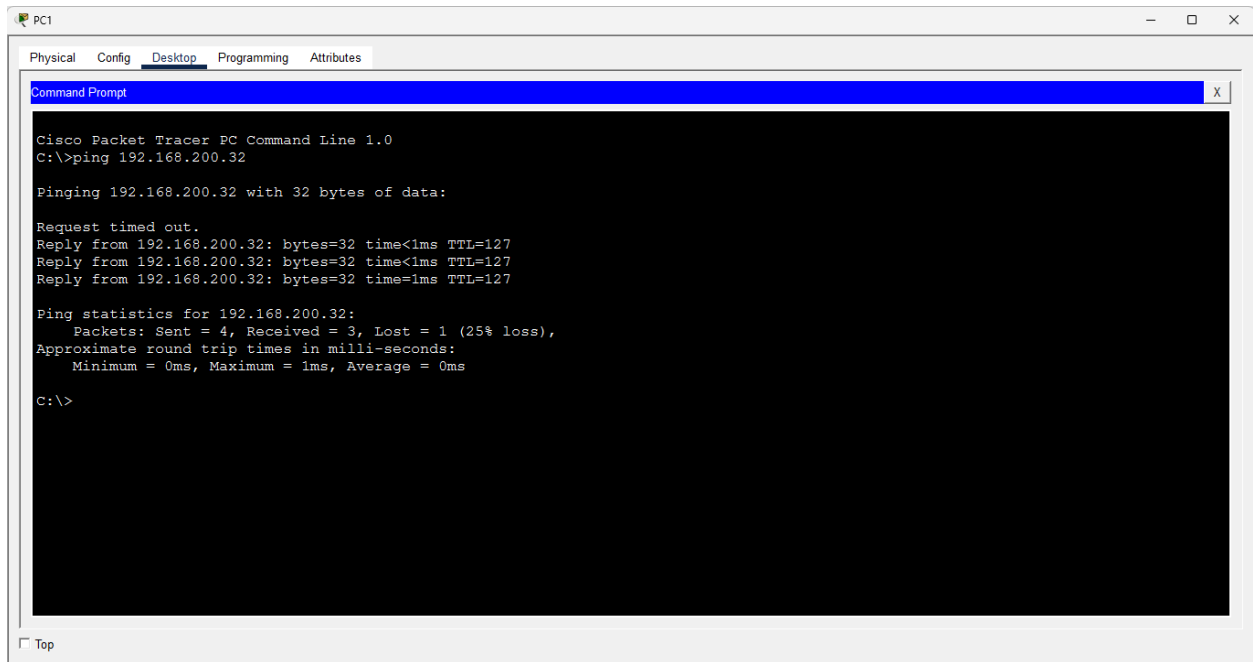
```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int gi0/1.100
R1(config-subif)#no sh
R1(config-subif)#en
R1(config-subif)#encapsulation do
R1(config-subif)#encapsulation dot1Q vlan 100
^
% Invalid input detected at '^' marker.

R1(config-subif)#encapsulation dot1Q 100
R1(config-subif)#ip ad
R1(config-subif)#ip address 192.168.100.1 255.255.255.0
R1(config-subif)#ex
R1(config)#int gi0/1.200
R1(config-subif)#no sh
R1(config-subif)#encapsulation dot1Q 200
R1(config-subif)#ip address 192.168.200.1 255.255.255.0
R1(config-subif)#ex
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

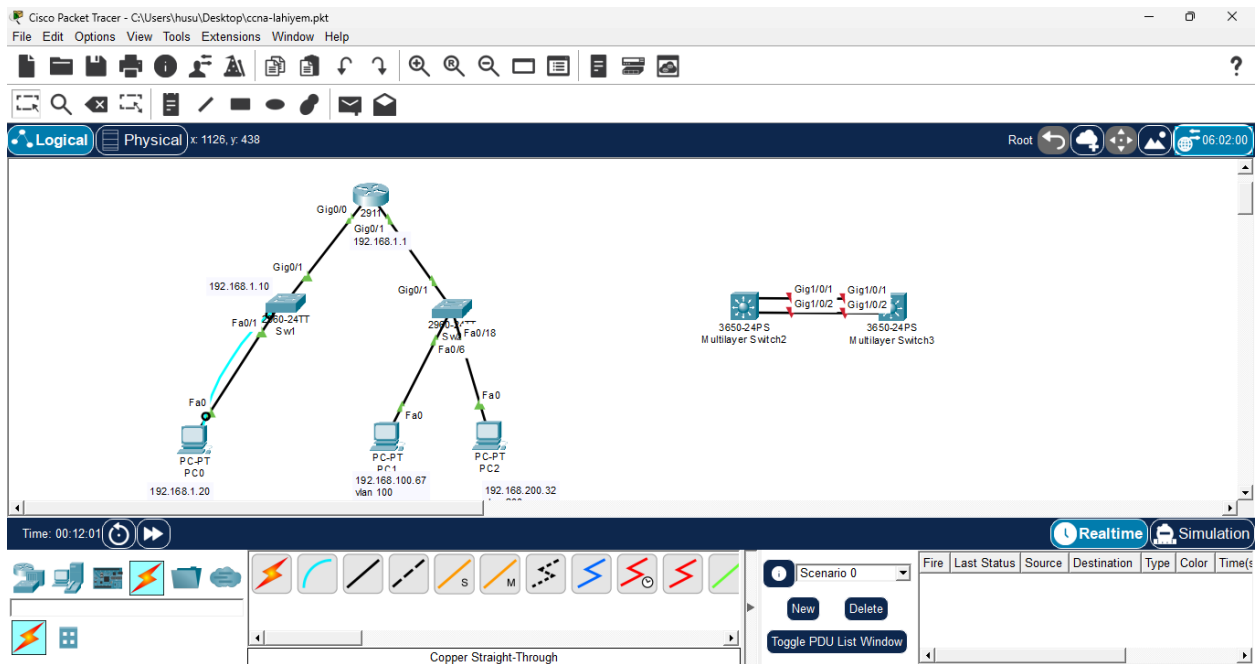
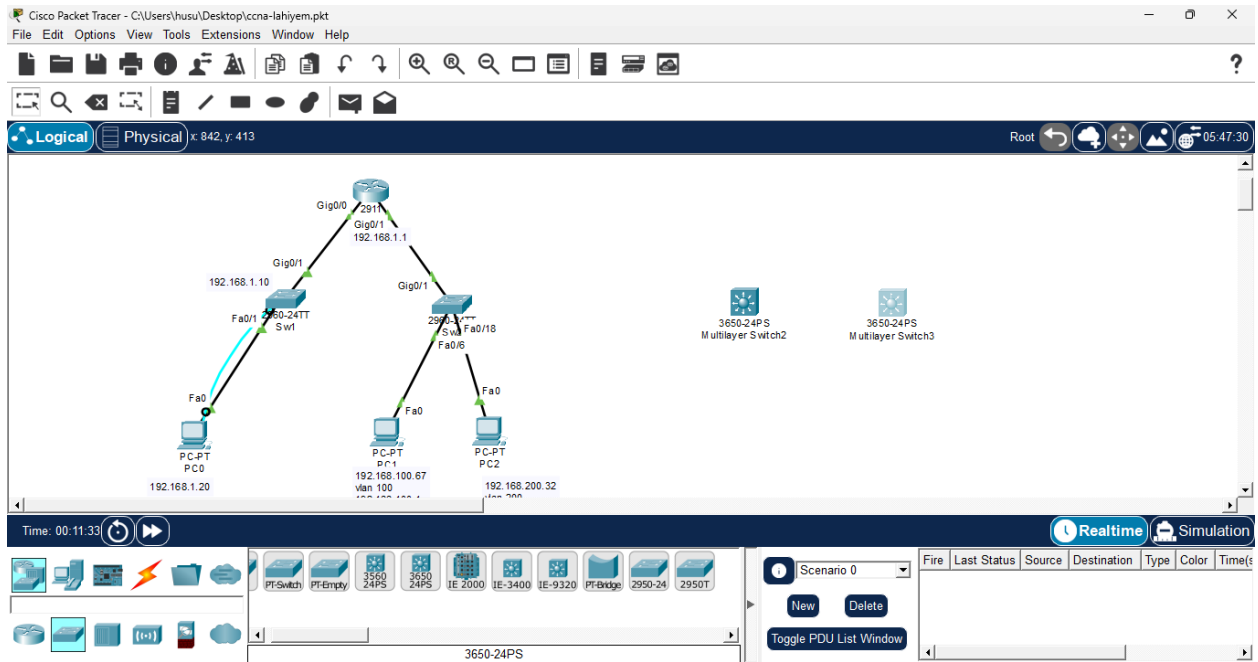
R1#wr
Building configuration...
[OK]
R1#
```

Copy Paste

Top



Multilayerde Etherchannelin qurulmasi ucun arxitektura



Multilayer Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```
Would you like to enter the initial configuration dialog? [yes/no]:
%LINK-5-CHANGED: Interface GigabitEthernet1/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet1/0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to up

Press RETURN to get started!

Switch>
Switch>
Switch>en
Switch#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int gi1/
Switch(config)#int gi1/0
Switch(config)#int re
Switch(config)#int ra
Switch(config)#int range gi
Switch(config)#int range gigabitEthernet 1/0/1-2
Switch(config-if-range)#
```

Copy Paste

Top

Multilayer Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch#en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int ra
Switch(config)#int range gi
Switch(config)#int range gigabitEthernet 1/0/1-2
Switch(config-if-range)#cha
Switch(config-if-range)#chane
Switch(config-if-range)#chanel-group 1 mode desirable
Switch(config-if-range)#channel-group 1 mode desirable
Switch(config-if-range)#channel-group 1 mode desirable
%EC-5-L3DONTENDL2: Gig1/0/1 suspended: PAGP currently not enabled on the remote port.
%EC-5-L3DONTENDL2: Gig1/0/2 suspended: PAGP currently not enabled on the remote port.
Switch(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to up

Switch(config-if-range)#
```

Copy Paste

Top

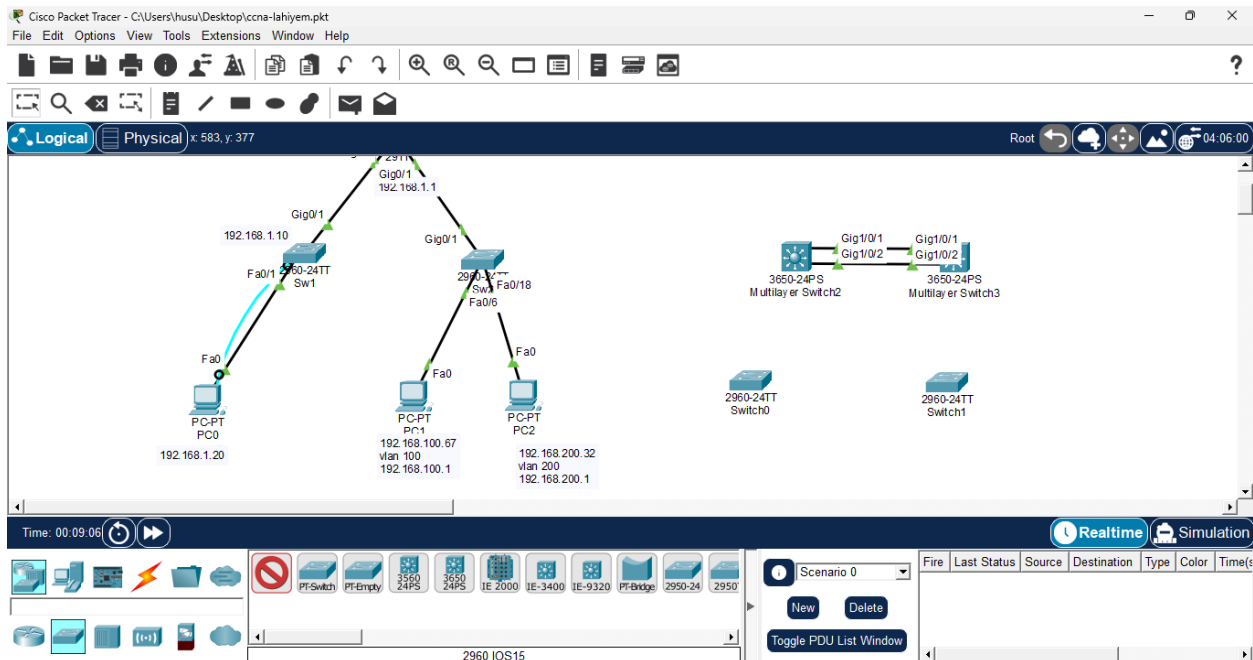
```
Multilayer Switch3
Physical Config CLI Attributes
IOS Command Line Interface

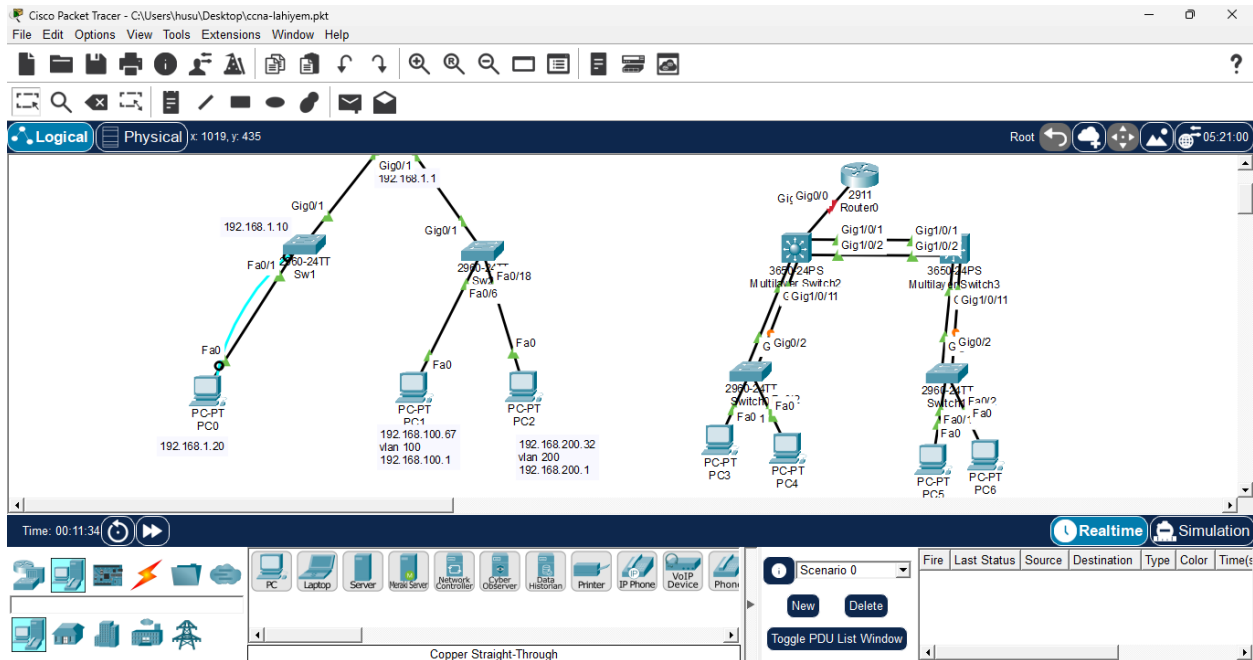
Switch>
Switch>en
Switch#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host DS1
DS1(config)#int ran
DS1(config)#int range gi1/0/1-2
DS1(config-if-range)#channel-group 1 mode desirable
%EC-5-L3DONTBNDL2: Gig1/0/1 suspended: PAGP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Gig1/0/2 suspended: PAGP currently not enabled on the remote port.
DS1(config-if-range)#
DS1(config)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to up
%LINK-5-CHANGED: Interface Port-channel1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

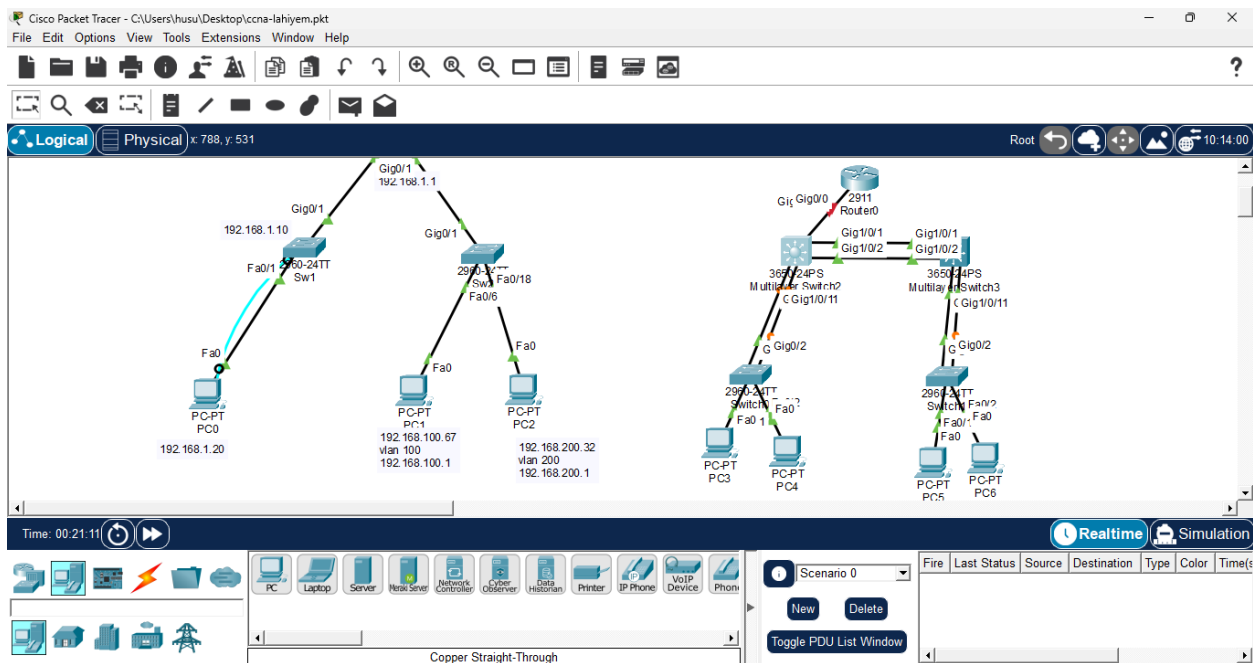
DS1(config-if-range)#^Z
```

Advanced Etherchannel arxitekturanin qurulmasi





Advanced Etherchannel arxitekturasinin konfigurasiyasi



Multilayer Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/11, changed state to up

DS1>
DS1>en
DS1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
DS1(config)#int range gig1/0/10-11
DS1(config-if-range)#chan
DS1(config-if-range)#chann
DS1(config-if-range)#channel-
DS1(config-if-range)#channel-group mode ?
% Unrecognized command
DS1(config-if-range)#channel-group 2 mode ?
    active      Enable LACP unconditionally
    auto        Enable PAGP only if a PAGP device is detected
    desirable   Enable PAGP unconditionally
    on          Enable Etherchannel only
    passive     Enable LACP only if a LACP device is detected
DS1(config-if-range)#channel-group 2 mode des
DS1(config-if-range)#channel-group 2 mode desirable
%EC-5-L3DONTBNDL2: Gig1/0/10 suspended: PAGP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Gig1/0/11 suspended: PAGP currently not enabled on the remote port.
DS1(config-if-range)#
Creating a port-channel interface Port-channel 2

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/10, changed state to down
```

Copy Paste

Top

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int gi 0/1-2
    ^
% Invalid input detected at '^' marker.

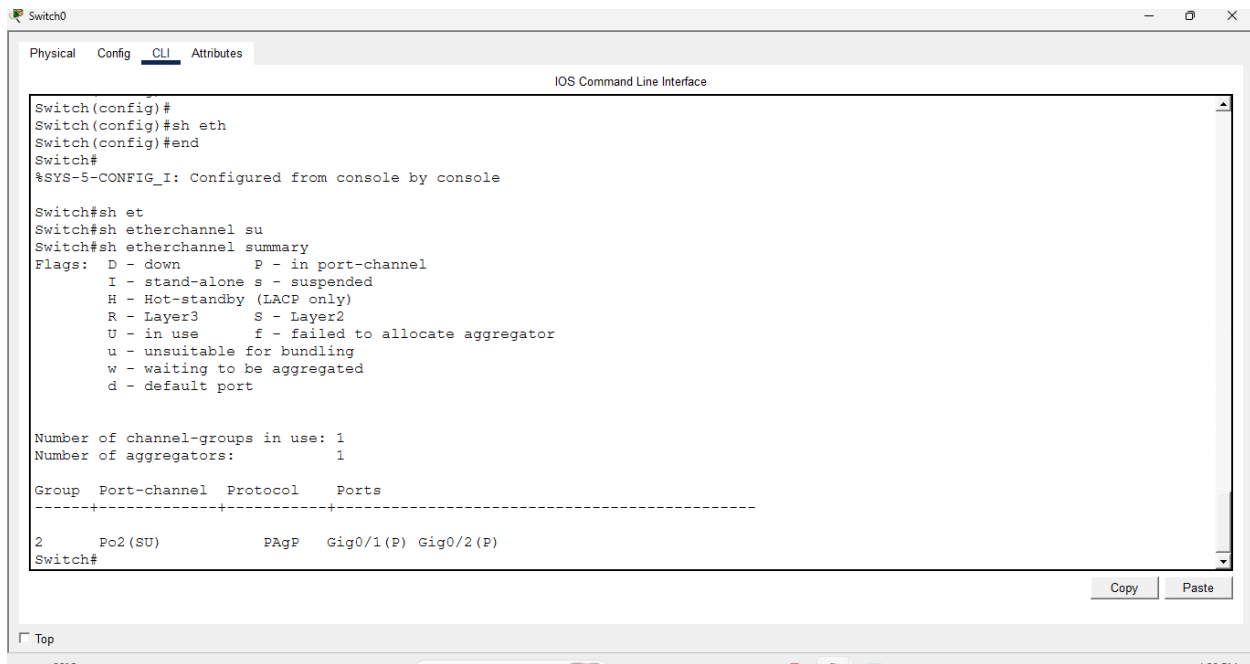
Switch(config)#int range gi 0/1-2
Switch(config-if-range)#chann
Switch(config-if-range)#channel-group 2 mode des
Switch(config-if-range)#channel-group 2 mode desirable
%EC-5-L3DONTBNDL2: Gig0/1 suspended: PAGP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Gig0/2 suspended: PAGP currently not enabled on the remote port.
Switch(config-if-range)#
Creating a port-channel interface Port-channel 2

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface Port-channel2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to up

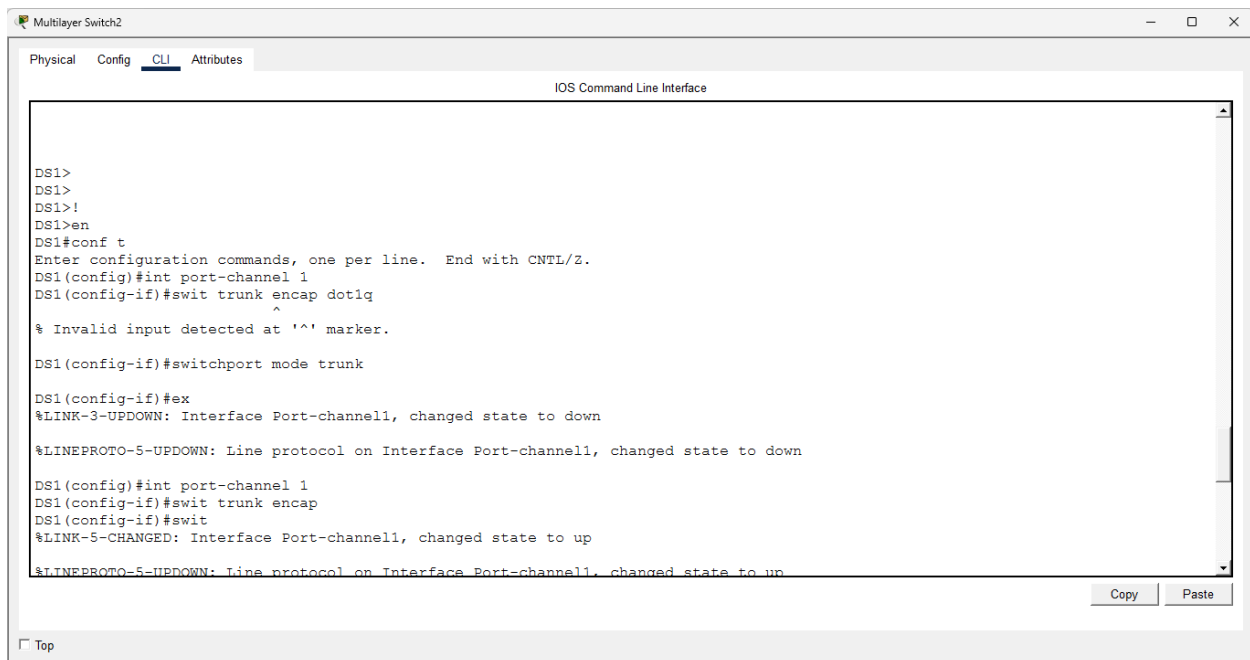
Switch(config-if-range)#
```

Copy Paste

Top



Switclerde trunkin konfigurasiyasi



Multilayer Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```
native          Set trunking native characteristics when interface is in
                  trunking mode
DS1(config-if)#switchport mode
DS1(config-if)#switchport mode tr
DS1(config-if)#switchport mode trunk
DS1(config-if)#
%LINK-3-UPDOWN: Interface Port-channel1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to down

DS1(config-if)#
DS1(config-if)#do sh int trunk
%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

Port      Mode      Encapsulation  Status      Native vlan
Po1       on        802.1q         trunking    1

Port      Vlans allowed on trunk
Po1       1-1005

Port      Vlans allowed and active in management domain
Po1       1

Port      Vlans in spanning tree forwarding state and not pruned
Po1       none
```

Copy Paste

Top

Multilayer Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```
DS1(config-if)#
DS1(config-if)#
DS1(config-if)#
DS1(config-if)#ex
DS1(config)#int port-channel 2
DS1(config-if)#switchport mode trunk

DS1(config-if)#!
%LINK-3-UPDOWN: Interface Port-channel2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to down

DS1(config-if)#
%LINK-5-CHANGED: Interface Port-channel2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to up

DS1(config-if)#switchport mode trunk
DS1(config-if)#
%LINK-3-UPDOWN: Interface Port-channel2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to down

DS1(config-if)#
DS1(config-if)#ex
DS1(config)#
```

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Top

Switch0

Physical Config **CLI** Attributes

IOS Command Line Interface

```
0 00 Gig0/1 Desirable-S1 0
0 00 Gig0/2 Desirable-S1 0
Time since last port bundled: 00d:00h:24m:59s Gig0/2

Switch# %SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk Port-channel2 VLAN1.
%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking Port-channel2 on VLAN0001. Inconsistent port type.

Switch#
Switch#
Switch#en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int port-channel 2
Switch(config-if)#switchport mode trunk

Switch(config-if)#!
%LINK-3-UPDOWN: Interface Port-channel2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to down

%LINK-5-CHANGED: Interface Port-channel2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to up

Switch(config-if)#
```

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Top

Multilayer Switch3

Physical Config **CLI** Attributes

IOS Command Line Interface

```
%SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking Port-channel1 on VLAN0001. Port consistency restored.
%SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking Port-channel1 on VLAN0001. Port consistency restored.

DS1>!
DS1>en
DS1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
DS1(config)#int port-channel 1
DS1(config-if)#switchport mode trunk
DS1(config-if)#ex
DS1(config)#int port-channel 3
DS1(config-if)#switchport mode trunk

DS1(config-if)#!
%LINK-3-UPDOWN: Interface Port-channel1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to down

%LINK-3-UPDOWN: Interface Port-channel3, changed state to down

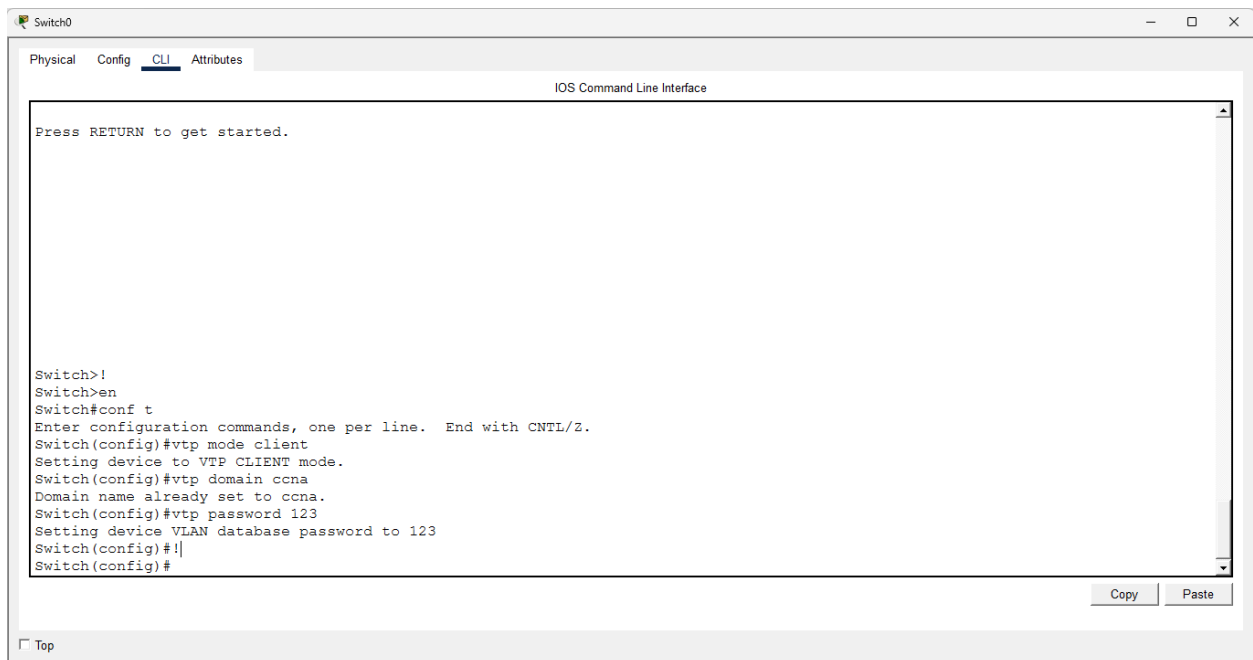
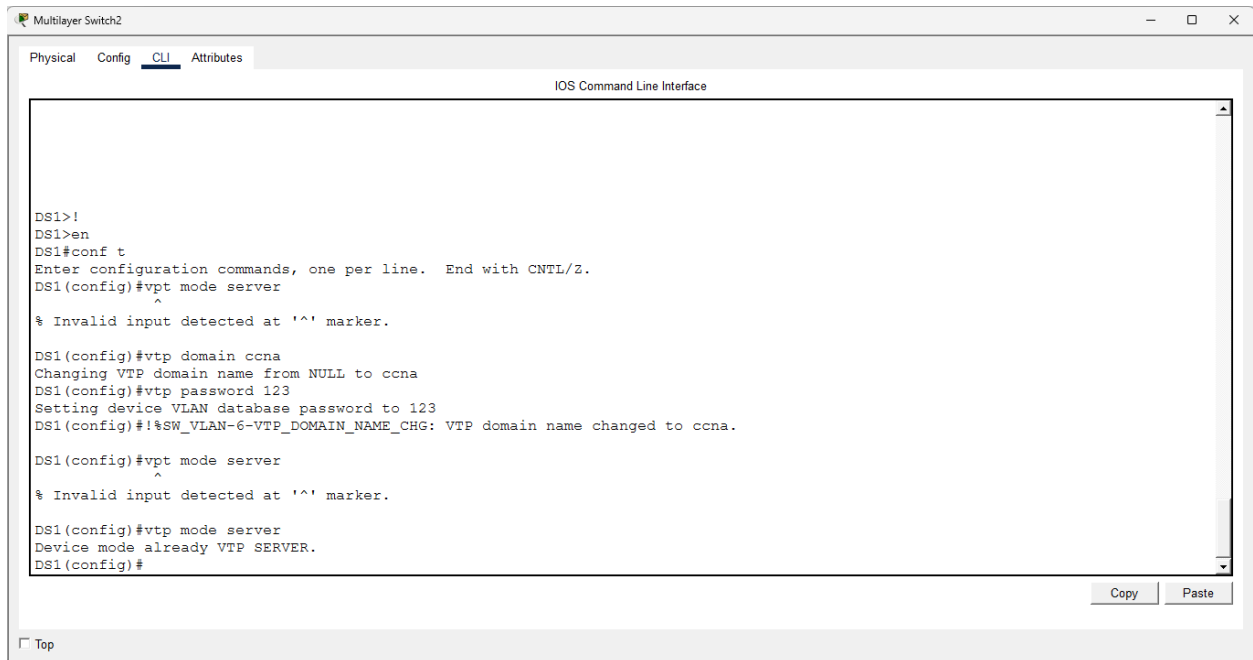
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel3, changed state to down

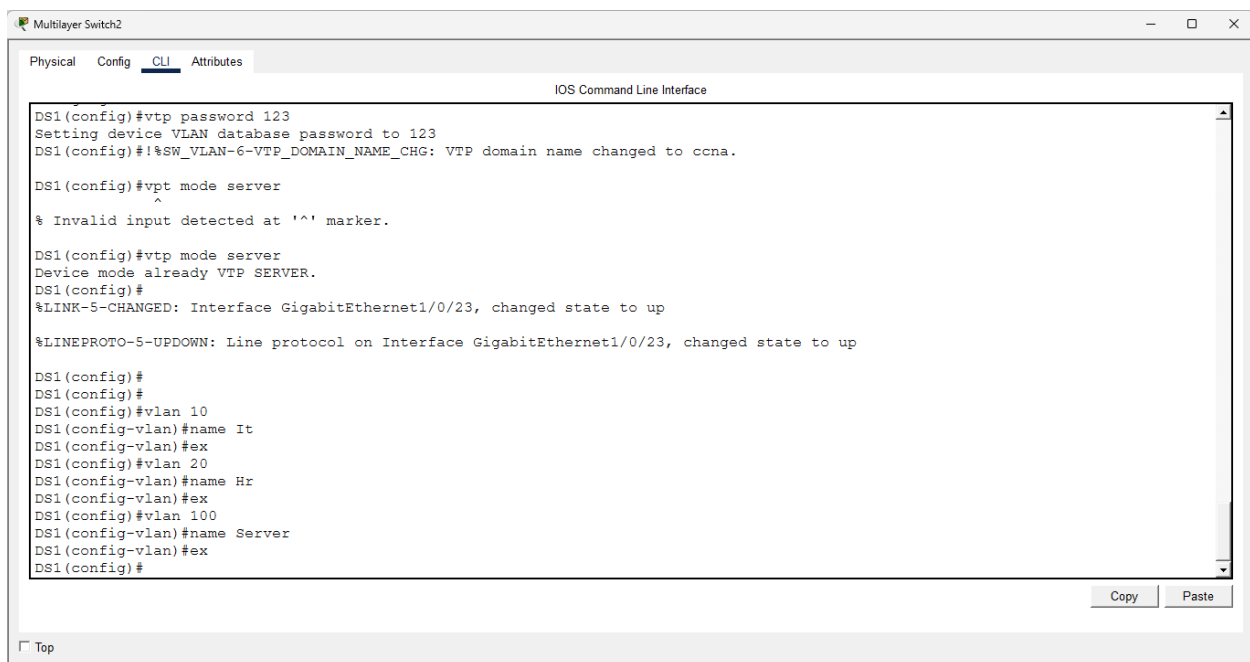
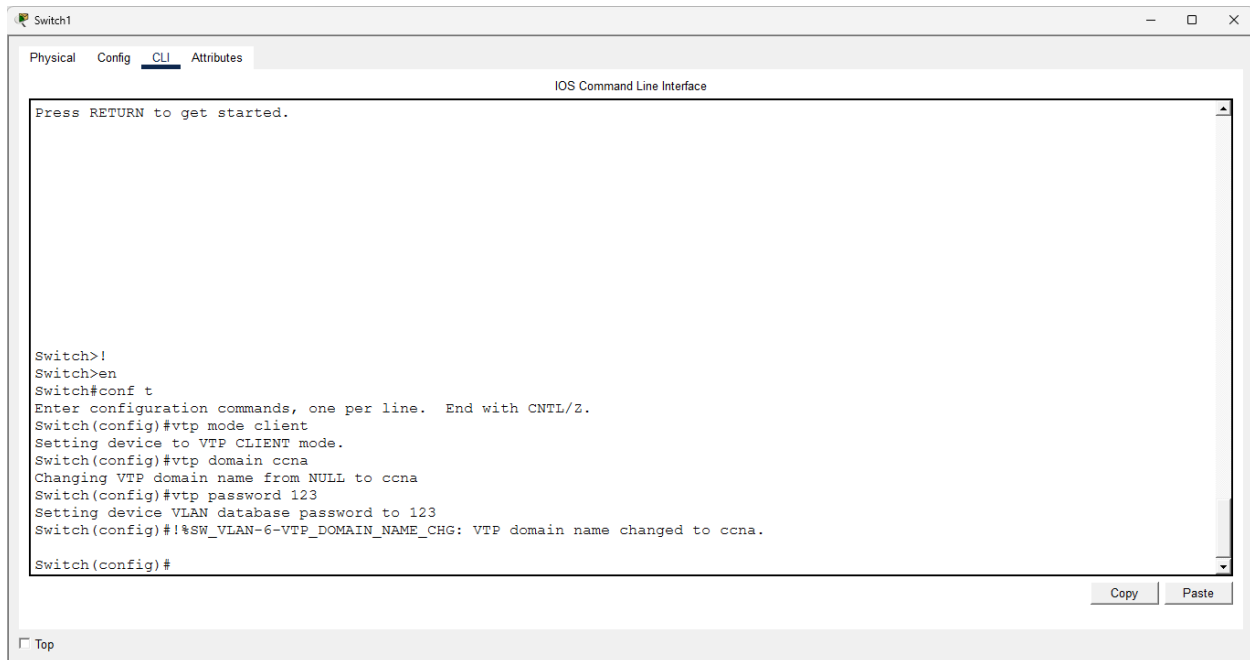
DS1(config-if)#
```

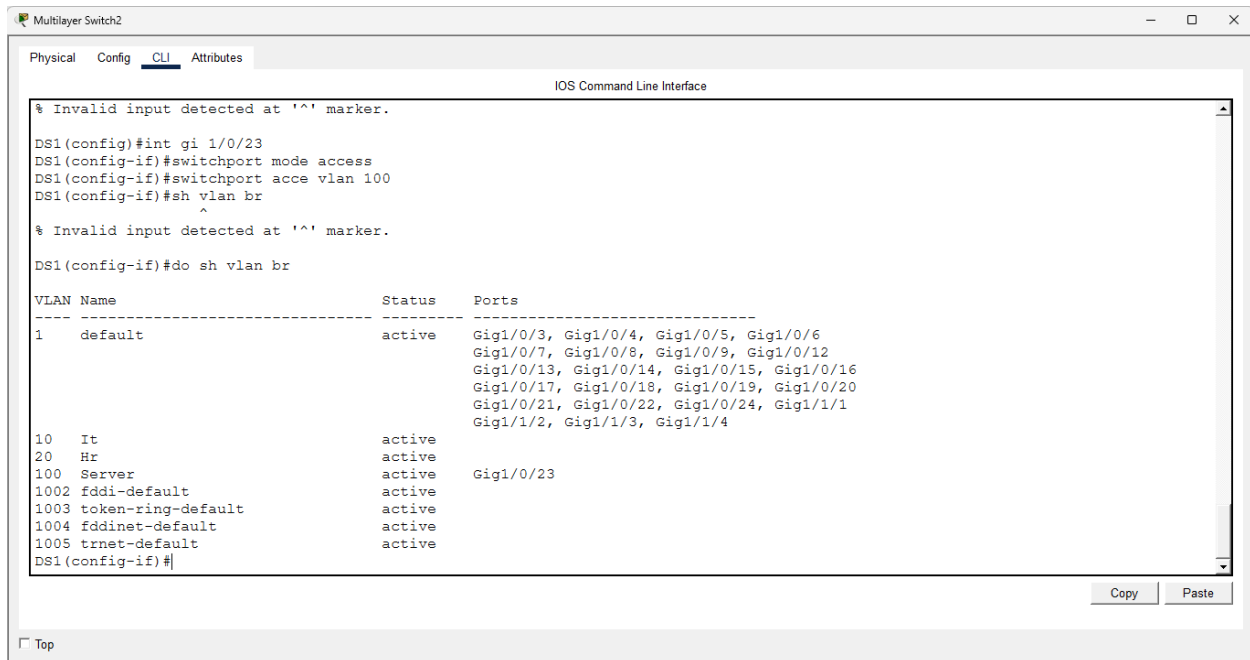
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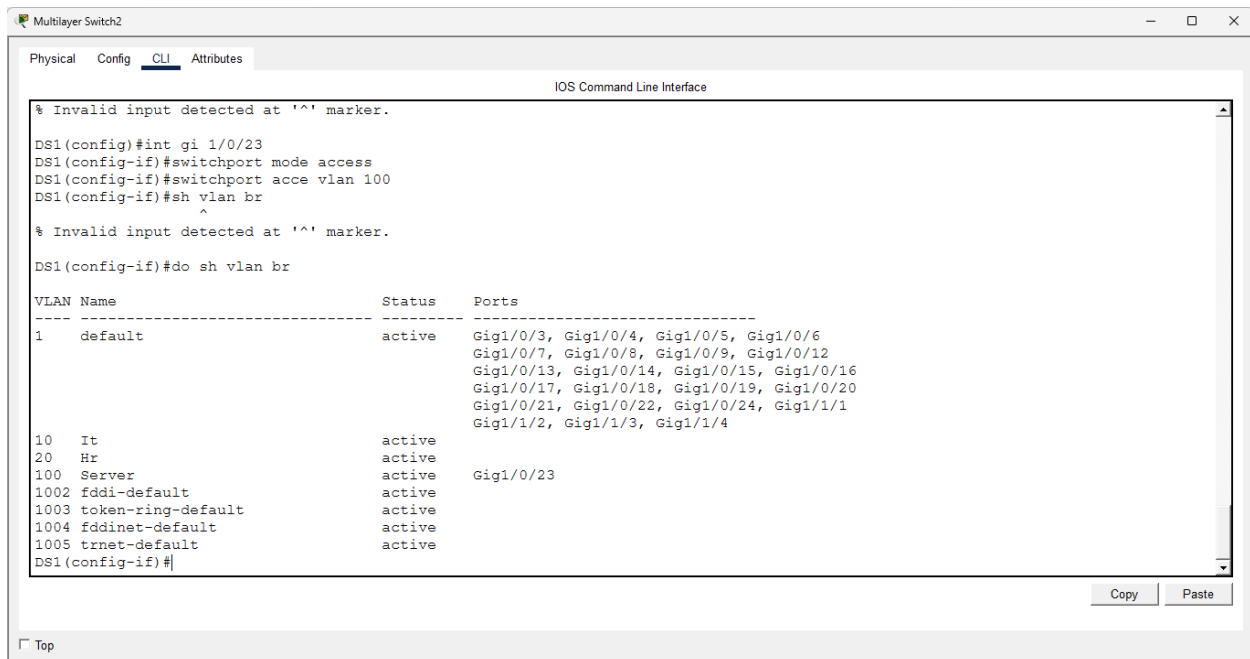
Vtp kofiqurasiyasi







Vtpden gelen vlanlarin switchlerin portuna qosulmasi



Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch(config-if)#no switchport acce vlan 100
Switch(config-if)#
%EC-5-CANNOT_BUNDLE2: Gig0/1 is not compatible with Po2 and will be suspended (dtp mode of Gig0/1 is off, Po2 is on)

Switch(config-if)#
%EC-5-CANNOT_BUNDLE2: Gig0/1 is not compatible with Po2 and will be suspended (dtp mode of Gig0/1 is off, Po2 is on)

%LINK-3-UPDOWN: Interface Port-channel2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to down

Switch(config-if)#
Switch(config-if)#
Switch(config-if)#ex
Switch(config)#
Switch(config)#!
Switch(config)#en
% Ambiguous command: "en"
Switch(config)#conf t
^
% Invalid input detected at '^' marker.

Switch(config)#int fa 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport acce vlan 10
Switch(config-if)#!
Switch(config-if)#
```

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Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>
Switch>!
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport acce vlan 20
Switch(config-if)#!
Switch(config-if)#
Switch(config-if)#!
Switch(config-if)#en
% Ambiguous command: "en"
Switch(config)#conf t
^
% Invalid input detected at '^' marker.

Switch(config)#int fa 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport acce vlan 10
Switch(config-if)#!
Switch(config-if)#
```

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Dhcp serverin konfigurasiyasi ucun vlanlarin konfigurasiyasi

Server0

Physical Config Services Desktop Programming Attributes

IP Configuration X

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.1.100.100

Subnet Mask 255.255.255.0

Default Gateway 10.1.100.1

DNS Server 8.8.8.8

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::203:E4FF:FE4B:3090

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Top

Multilayer Switch2

Physical Config CLI Attributes

IOS Command Line Interface

Enter configuration commands, one per line. End with CNTL/Z.

```
DS1(config)#int vlan 10
DS1(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

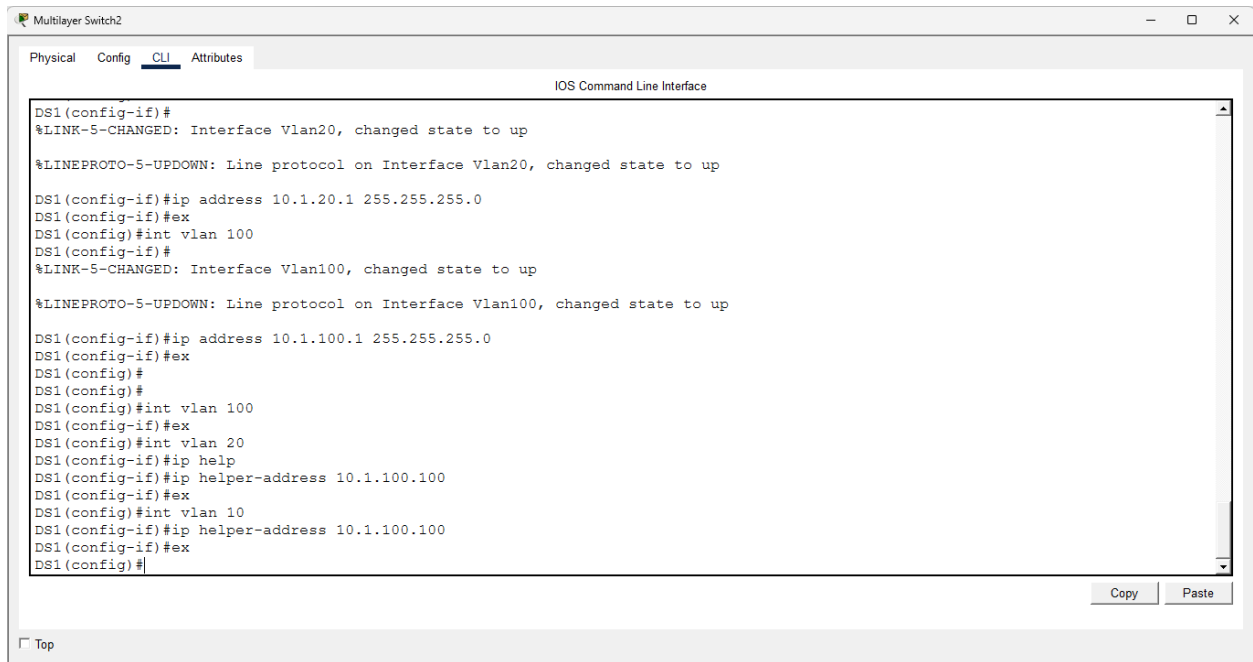
DS1(config-if)#ip add
DS1(config-if)#ip address 10.1.10.1 255.255.255.0
DS1(config-if)#ex
DS1(config)#int vlan 20
DS1(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

DS1(config-if)#ip address 10.1.20.1 255.255.255.0
DS1(config-if)#ex
DS1(config)#int vlan 100
DS1(config-if)#
%LINK-5-CHANGED: Interface Vlan100, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan100, changed state to up

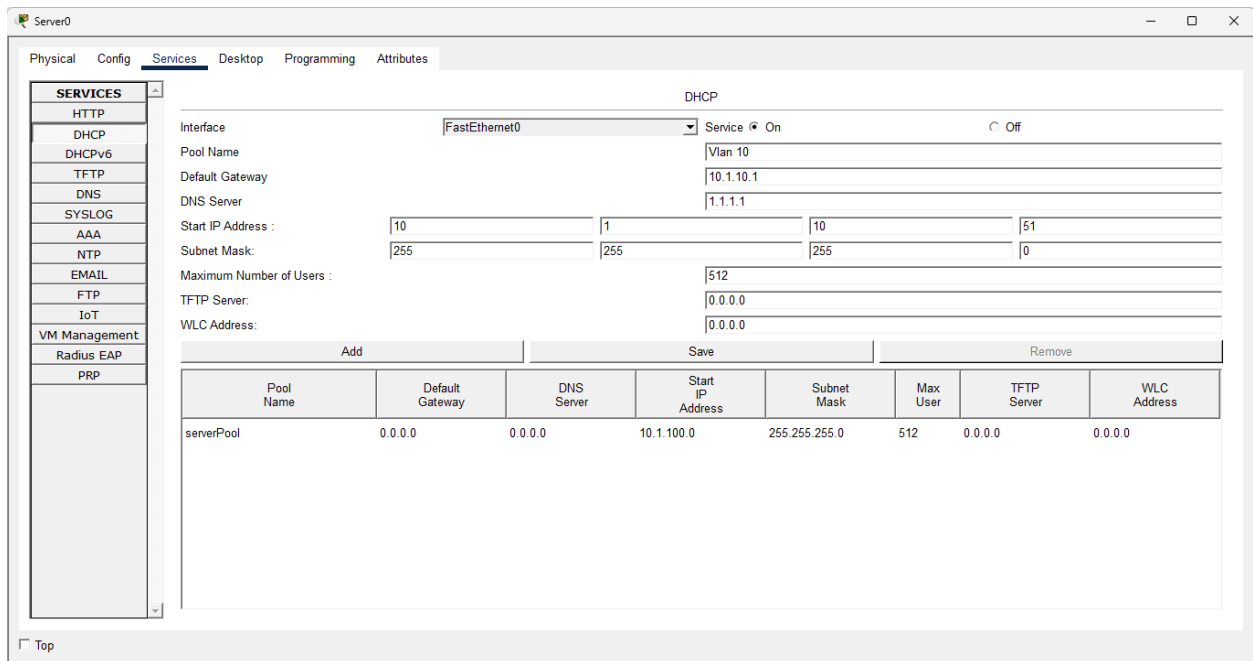
DS1(config-if)#ip address 10.1.100.1 255.255.255.0
DS1(config-if)#ex
DS1(config)#
```

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Vlanlar ucun dhcp poollarinin yaradilmasi



Server0

Physical
Config
Services
Desktop
Programming
Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

DHCP

Interface
FastEthernet0
Service
On
Off

Pool Name
Vlan 20

Default Gateway
10.1.20.1

DNS Server
1.1.1.1

Start IP Address :
10
1
20
51

Subnet Mask:
255
255
255
0

Maximum Number of Users :
205

TFTP Server:
0.0.0.0

WLC Address:
0.0.0.0

Add
Save
Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Vlan 20	10.1.20.1	1.1.1.1	10.1.20.51	255.255.255.0	205	0.0.0.0	0.0.0.0
Vlan 10	10.1.10.1	1.1.1.1	10.1.10.51	255.255.255.0	205	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	10.1.100.0	255.255.255.0	512	0.0.0.0	0.0.0.0

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PC3

Physical
Config
Desktop
Programming
Attributes

IP Configuration

Interface
FastEthernet0

IP Configuration

DHCP
Static

IPv4 Address
10.1.10.51

Subnet Mask
255.255.255.0

Default Gateway
10.1.10.1

DNS Server
1.1.1.1

IPv6 Configuration

Automatic
Static

IPv6 Address

Link Local Address
FE80::203:E4FF:FEDB:AE0C

Default Gateway

DNS Server

802.1X

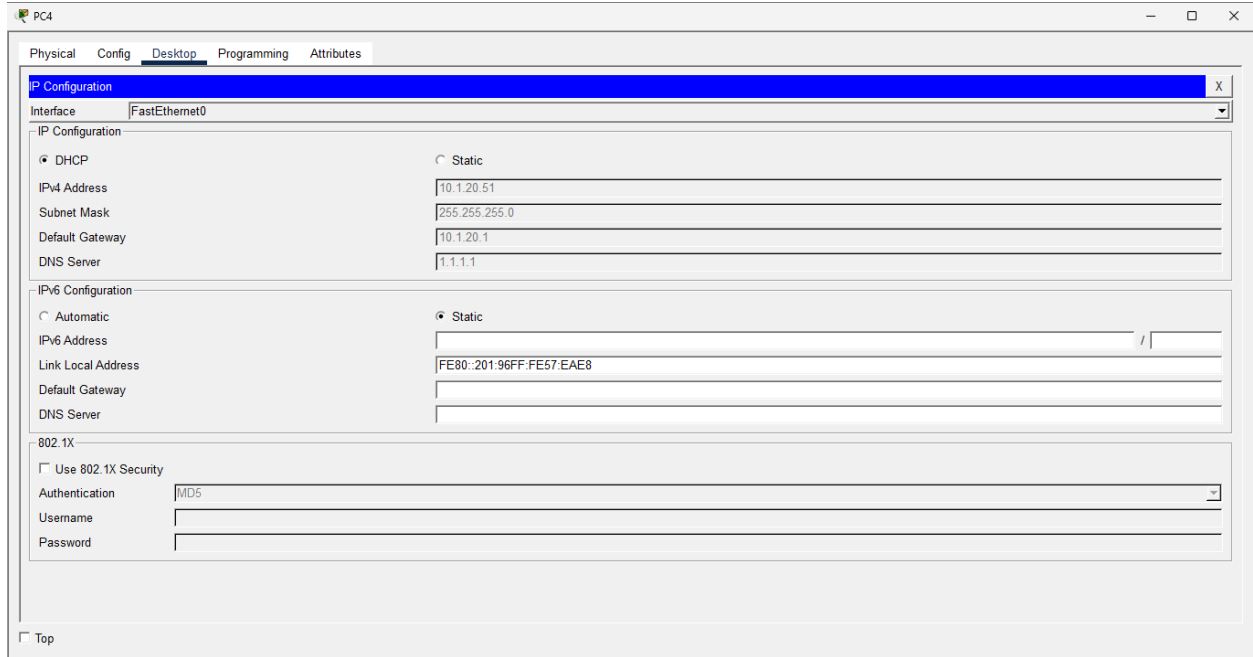
Use 802.1X Security

Authentication
MD5

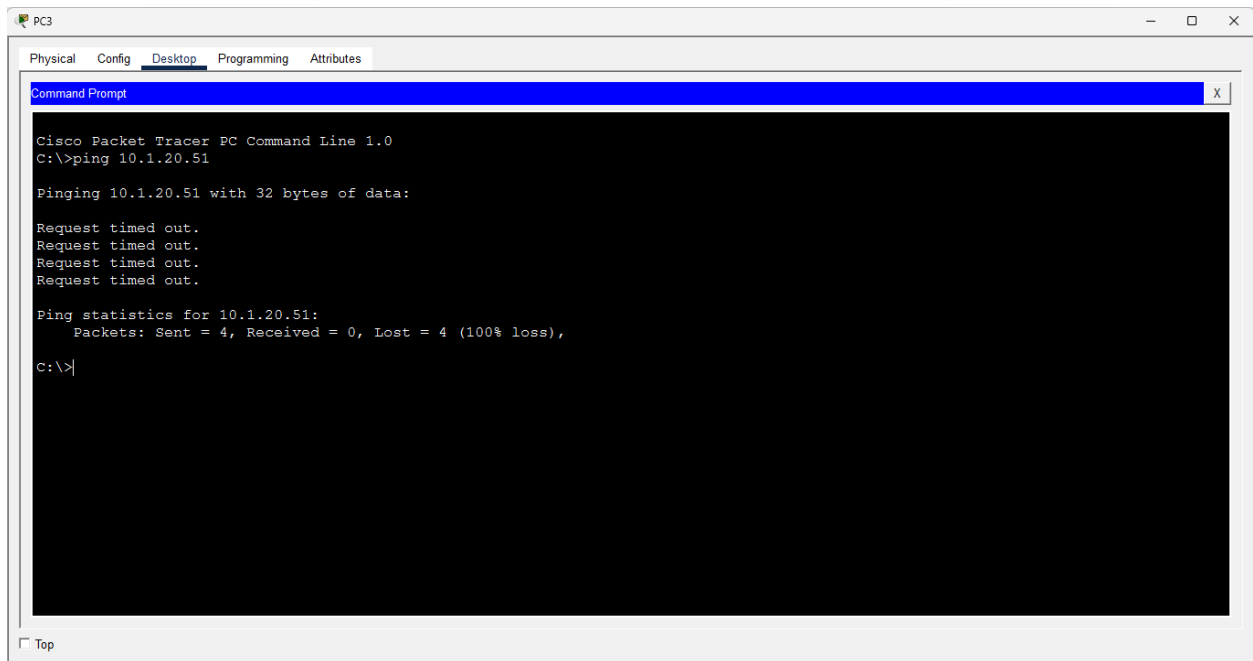
Username

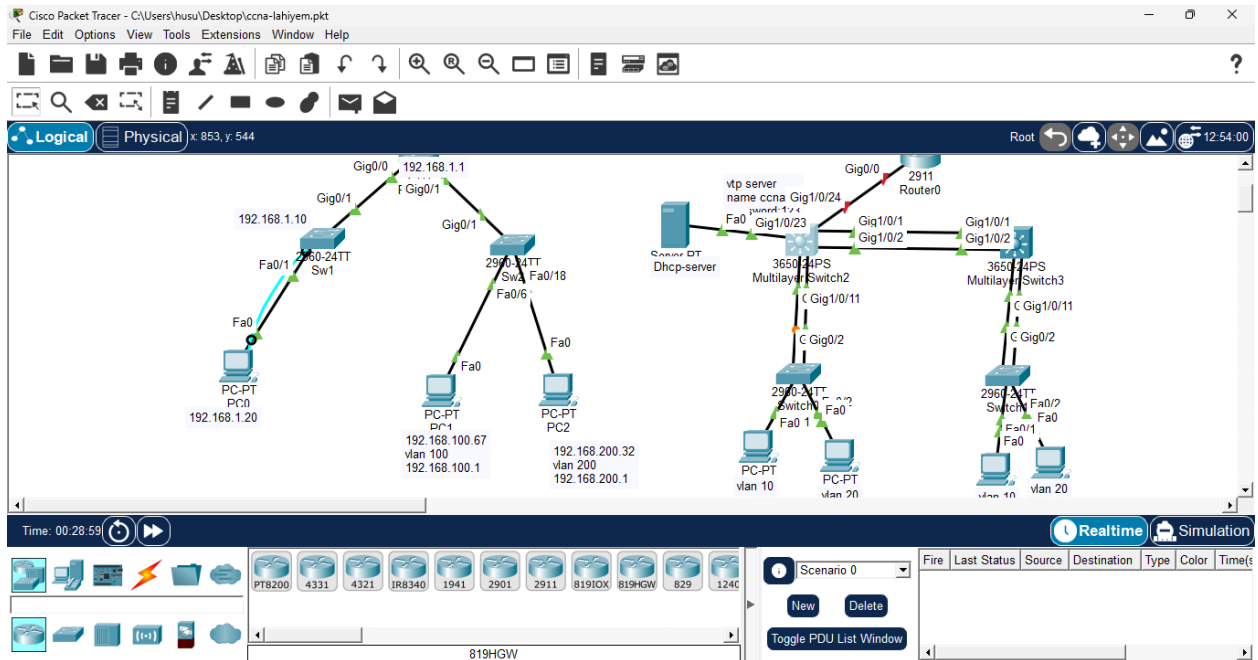
Password

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Vlanların routing edilmesi





Multilayer Switch2

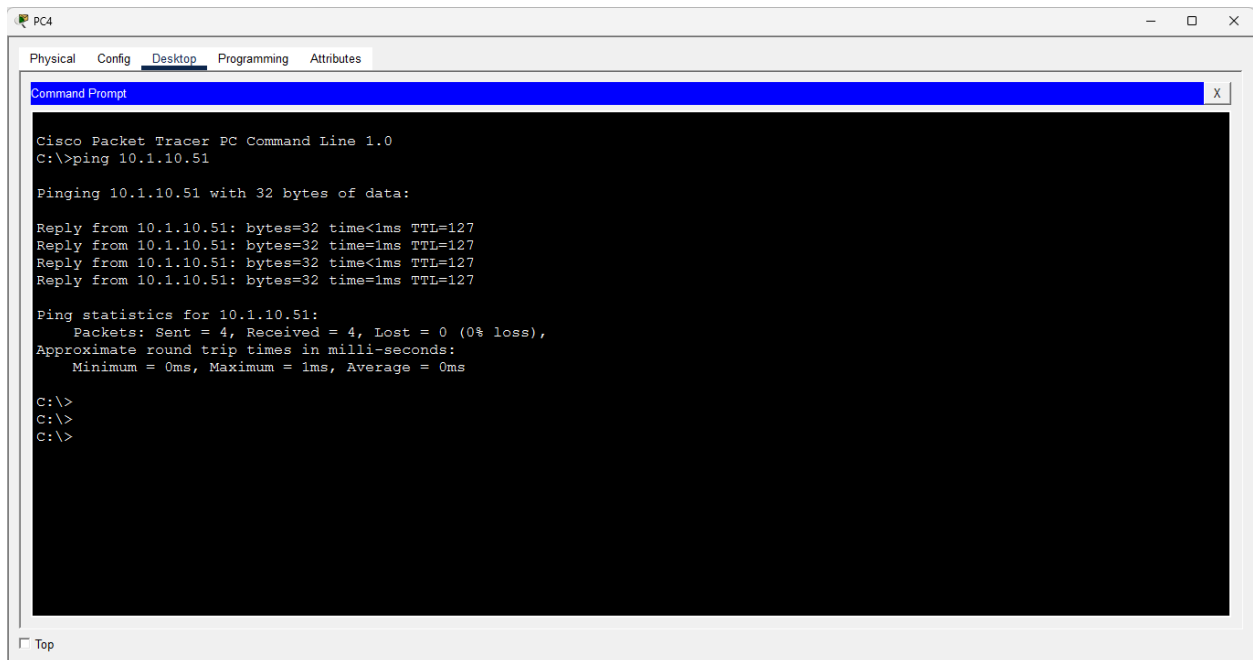
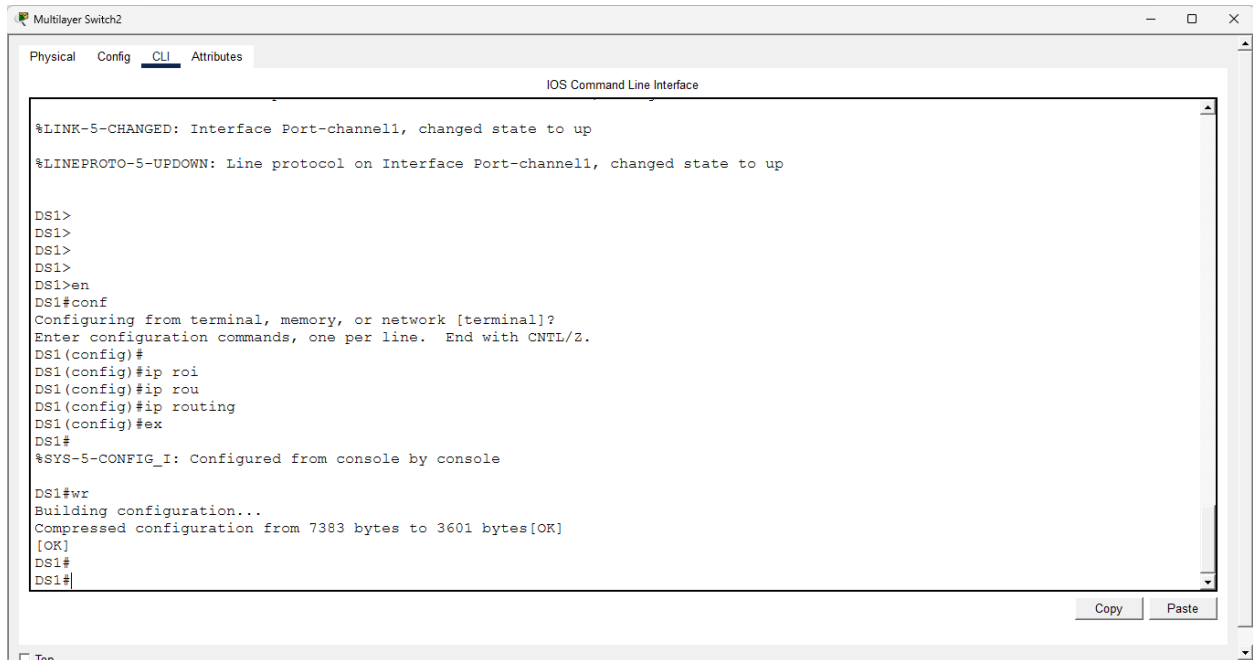
Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/11, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet1/0/23, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/23, changed state to up
%LINK-5-CHANGED: Interface Port-channel1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up
%LINK-3-UPDOWN: Interface Port-channel1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to down
%LINK-5-CHANGED: Interface Port-channel1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

DS1>
DS1>
DS1>
DS1>
DS1>en
DS1#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
DS1(config)#
```

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Spanning tree root primary ve secondary konfigurasiyasi

```
Multilayer Switch2
Physical Config CLI Attributes
IOS Command Line Interface

Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi1/0/10       Desg FWD 4          128.10 P2p
Gi1/0/11       Desg FWD 4          128.11 P2p
Po1            Desg FWD 3          128.29 P2p

VLAN0100
  Spanning tree enabled protocol ieee
  Root ID      Priority 32868
                Address 0001.C7CD.1CBB
                This bridge is the root
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

  Bridge ID    Priority 32868 (priority 32768 sys-id-ext 100)
                Address 0001.C7CD.1CBB
                Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
                Aging Time 20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi1/0/10       Desg FWD 4          128.10 P2p
Gi1/0/11       Desg FWD 4          128.11 P2p
Gi1/0/23       Desg FWD 19         128.23 P2p
Po1            Desg FWD 3          128.29 P2p

DS1#
```

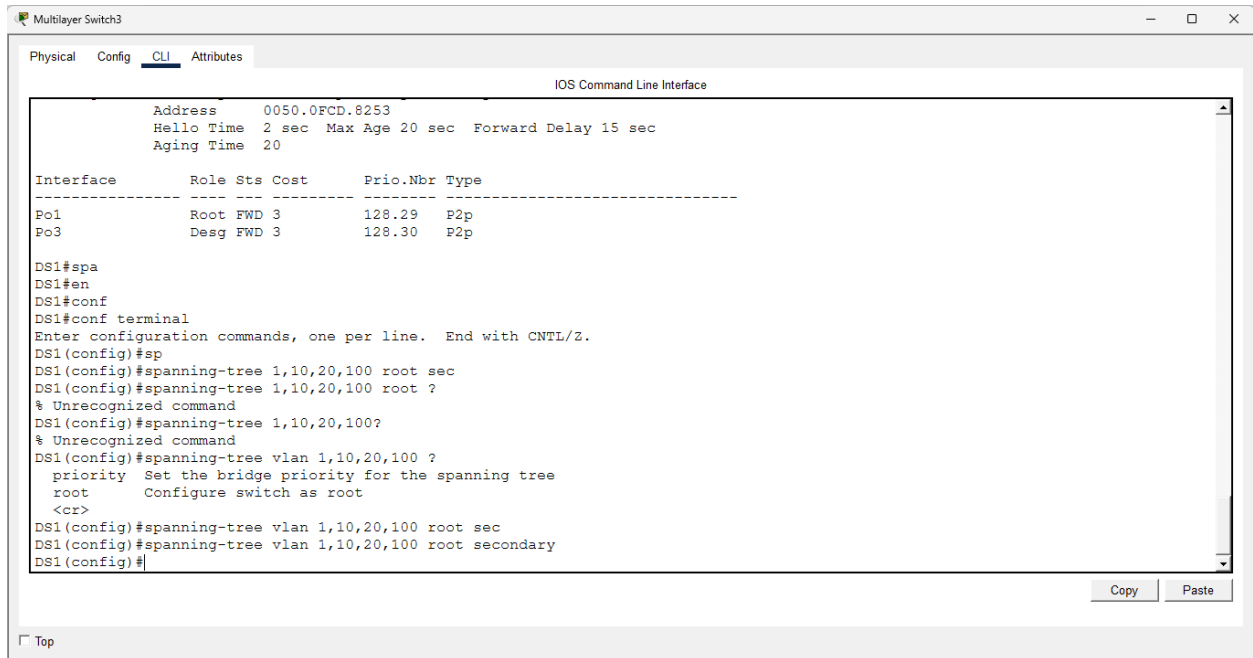
```
Multilayer Switch2
Physical Config CLI Attributes
IOS Command Line Interface

Root ID      Priority 32868
Address      0001.C7CD.1CBB
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

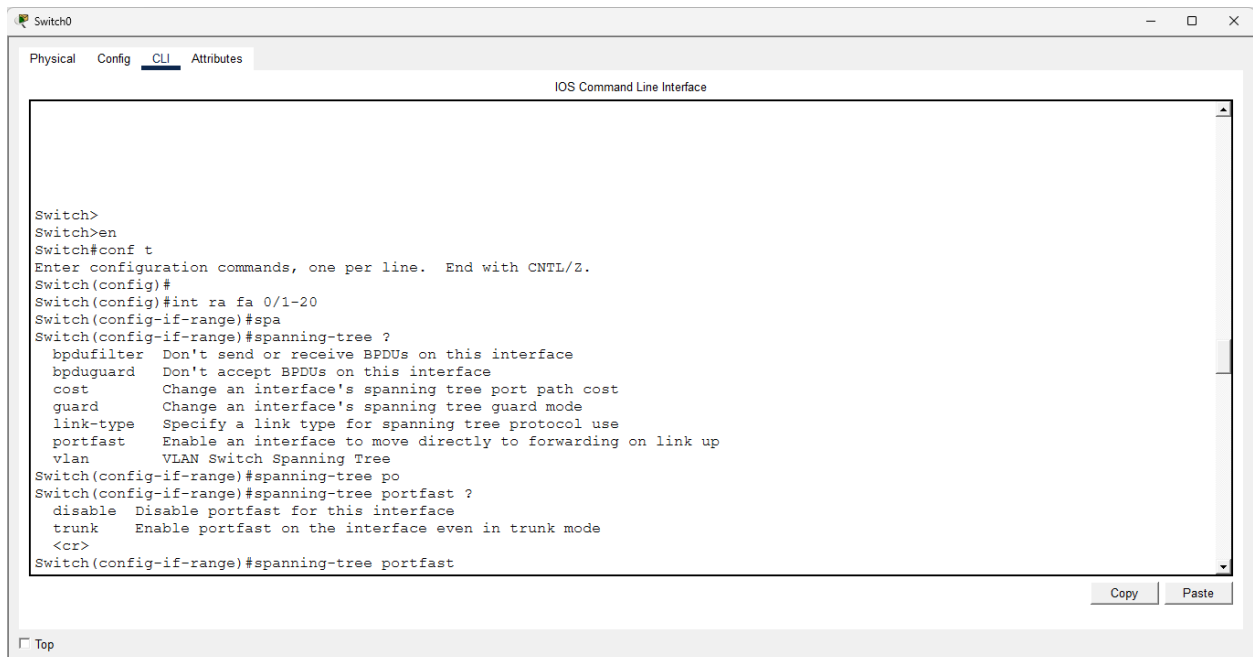
Bridge ID    Priority 32868 (priority 32768 sys-id-ext 100)
Address      0001.C7CD.1CBB
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

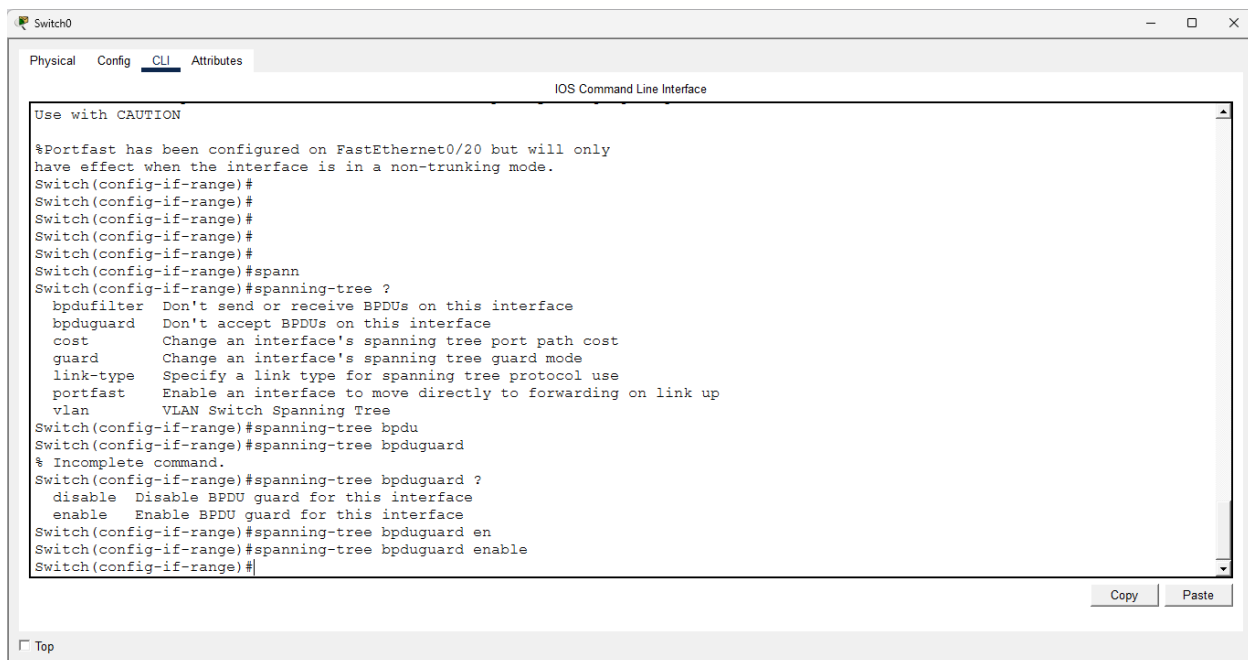
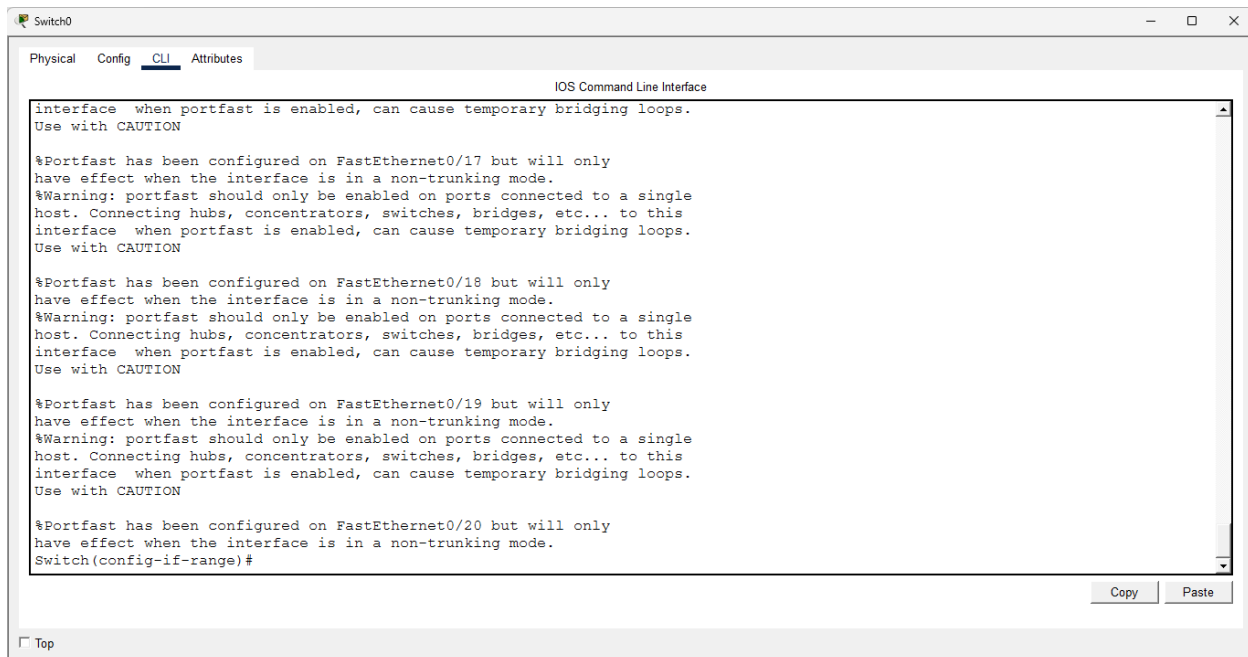
Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi1/0/10       Desg FWD 4          128.10 P2p
Gi1/0/11       Desg FWD 4          128.11 P2p
Gi1/0/23       Desg FWD 19         128.23 P2p
Po1            Desg FWD 3          128.29 P2p

DS1#en
DS1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
DS1(config)#spa
DS1(config)#spanning-tree vlan 1,10,20,100 pri
DS1(config)#spanning-tree vlan 1,10,20,100 prima
DS1(config)#spanning-tree vlan 1,10,20,100 roo
DS1(config)#spanning-tree vlan 1,10,20,100 root pri
DS1(config)#spanning-tree vlan 1,10,20,100 root primary
DS1(config)#
```



Spanning tree : Portfast ve bpduguard konfigurasiyasi





Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#
Switch(config)#int ra fa 0-
Switch(config)#int ra fa ?
<0-9> FastEthernet interface number
Switch(config)#int ra fa 0-20
^
% Invalid input detected at '^' marker.

Switch(config)#int ra fa 0/1-24
Switch(config-if-range)#
```

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Switch2

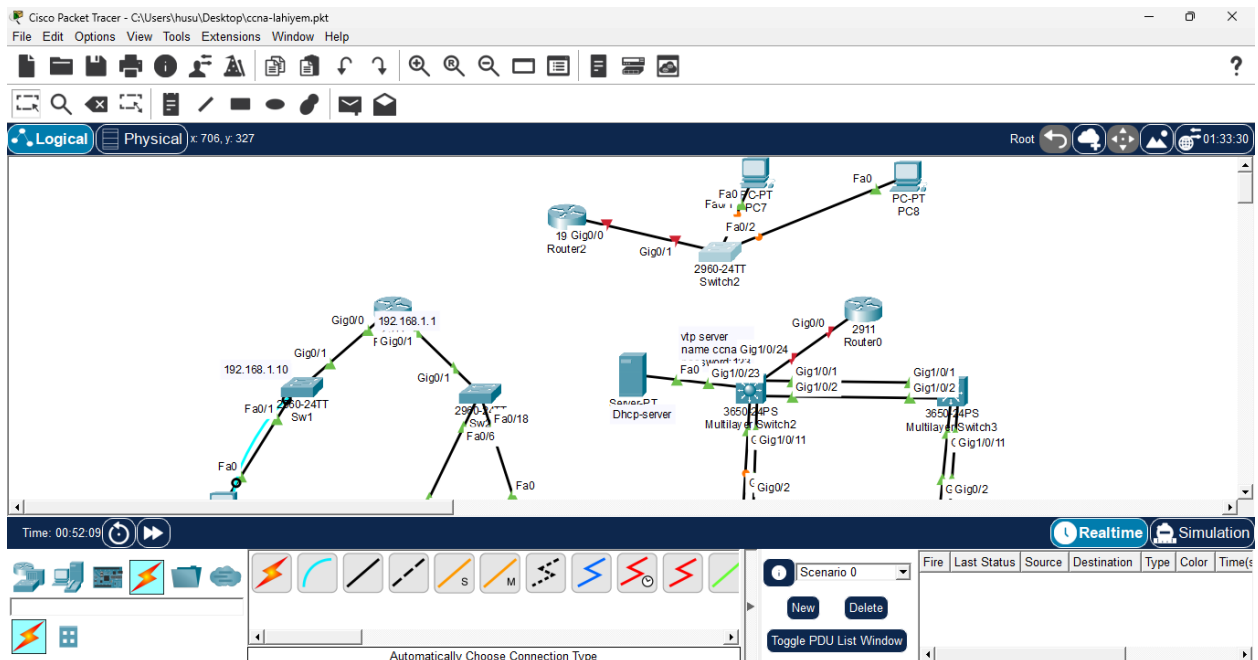
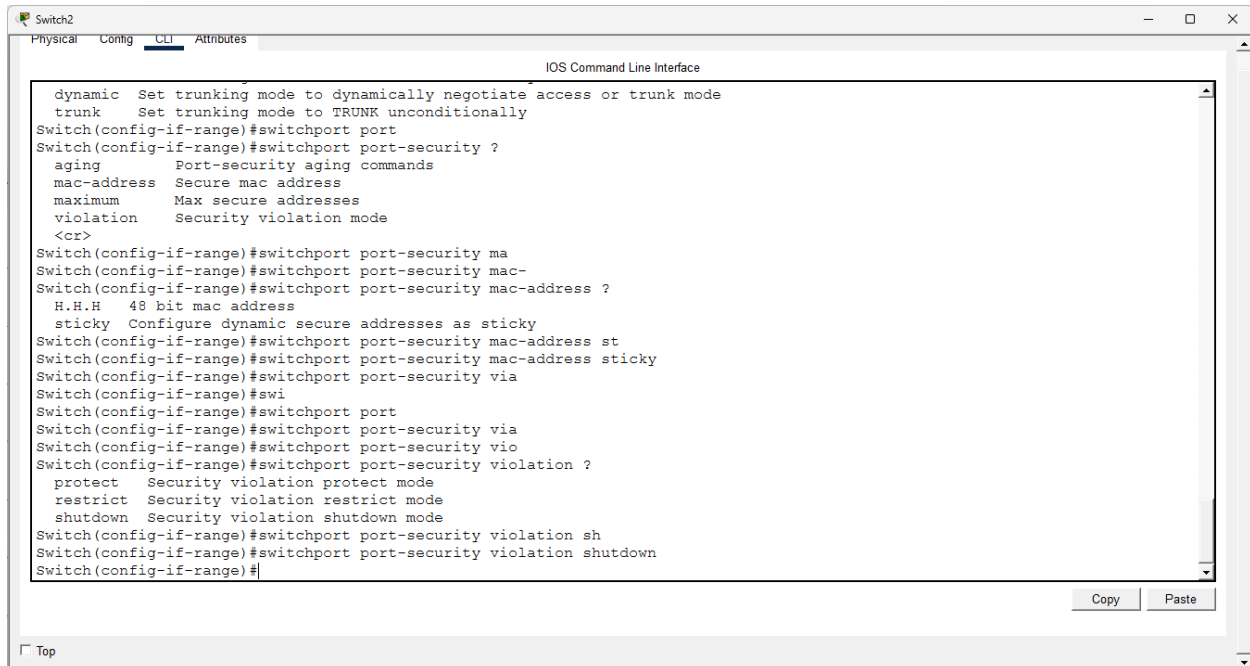
Physical Config CLI Attributes

IOS Command Line Interface

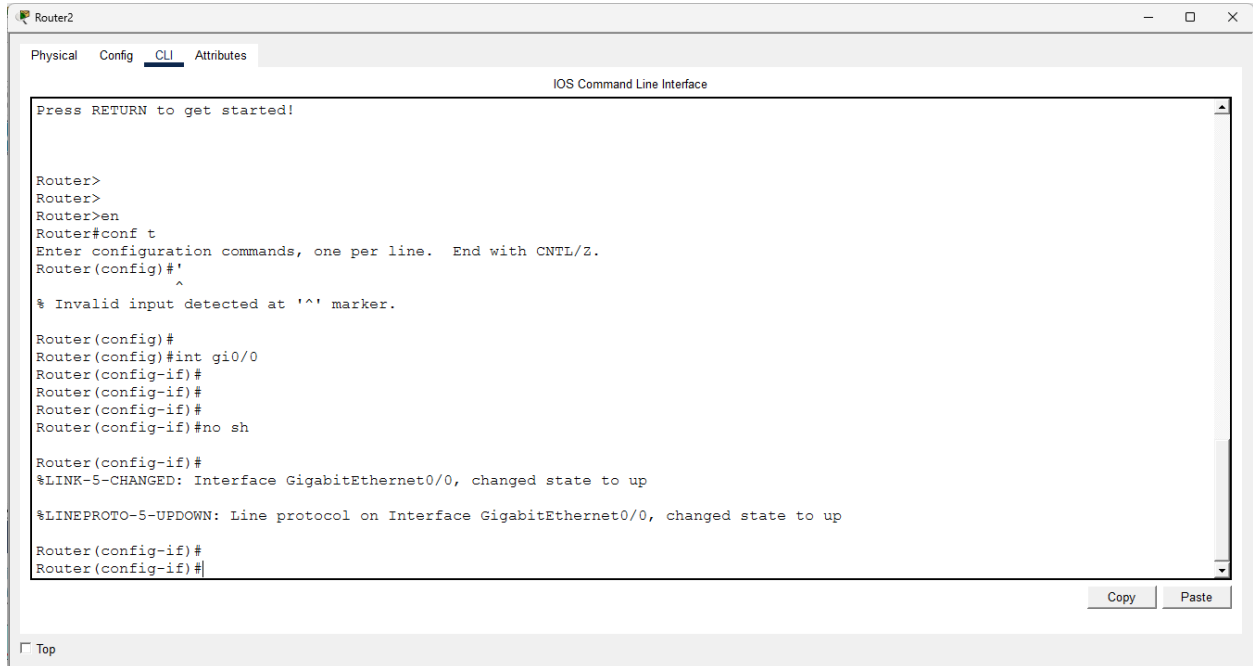
```
port-security interface
priority Security related command
priority Set appliance 802.1p priority
protected Configure an interface to be a protected port
trunk Set trunking characteristics of the interface
voice Voice appliance attributes
Switch(config-if-range)#switchport mod
Switch(config-if-range)#switchport mode
% Incomplete command.
Switch(config-if-range)#switchport mode ?
access Set trunking mode to ACCESS unconditionally
dynamic Set trunking mode to dynamically negotiate access or trunk mode
trunk Set trunking mode to TRUNK unconditionally
Switch(config-if-range)#switchport port
Switch(config-if-range)#switchport port-security ?
aging Port-security aging commands
mac-address Secure mac address
maximum Max secure addresses
violation Security violation mode
<Cr>
Switch(config-if-range)#switchport port-security ma
Switch(config-if-range)#switchport port-security mac-
Switch(config-if-range)#switchport port-security mac-address ?
H.H.H 48 bit mac address
sticky Configure dynamic secure addresses as sticky
Switch(config-if-range)#switchport port-security mac-address st
Switch(config-if-range)#switchport port-security mac-address sticky
Switch(config-if-range)#
```

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Routerda dhcp konfigurasiyasi



The screenshot shows the Router2 CLI interface with the 'CLI' tab selected. The command history shows the user entering 'en' to enter configuration mode, then 'conf t' to enter global configuration mode. They then enter 'int gi0/0' to select the interface, followed by 'no sh' to disable shutdown. The interface status changes to 'up'. The user then enters '^' to exit the configuration mode, which results in an 'Invalid input detected' error.

```
Router2
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started!

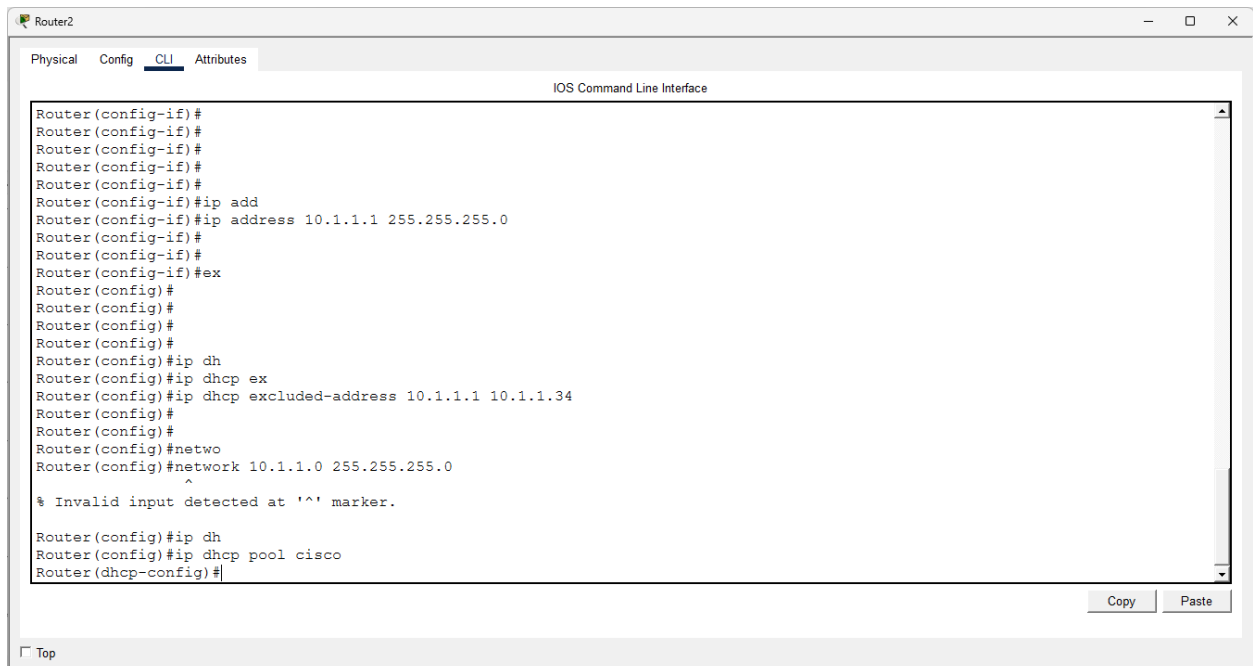
Router>
Router>
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#^
% Invalid input detected at '^' marker.

Router(config)#
Router(config)#int gi0/0
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#
```



The screenshot shows the Router2 CLI interface with the 'CLI' tab selected. The command history shows the user entering 'ip add' to start IP address configuration, then 'ip address 10.1.1.1 255.255.255.0' to assign the address. They then enter 'ex' to exit the configuration mode, which results in an 'Invalid input detected' error. The user then enters 'ip dh' to start DHCP configuration, followed by 'ip dhcp ex' to enter DHCP exclusion configuration mode. They then enter 'ip dhcp excluded-address 10.1.1.1 10.1.1.34' to define the excluded range. The user then enters '^' to exit the configuration mode, which results in an 'Invalid input detected' error. The user then enters 'ip dh' to start DHCP configuration, followed by 'ip dhcp pool cisco' to create a DHCP pool. The user then enters 'dhcp-config' to enter DHCP configuration mode.

```
Router2
Physical Config CLI Attributes
IOS Command Line Interface

Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#ip add
Router(config-if)#ip address 10.1.1.1 255.255.255.0
Router(config-if)#
Router(config-if)#
Router(config-if)#ex
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#ip dh
Router(config)#ip dhcp ex
Router(config)#ip dhcp excluded-address 10.1.1.1 10.1.1.34
Router(config)#
Router(config)#
Router(config)#netwo
Router(config)#network 10.1.1.0 255.255.255.0
Router(config)#^
% Invalid input detected at '^' marker.

Router(config)#ip dh
Router(config)#ip dhcp pool cisco
Router(dhcp-config)#
```

Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#ex
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#ip dh
Router(config)#ip dhcp ex
Router(config)#ip dhcp excluded-address 10.1.1.1 10.1.1.34
Router(config)#
Router(config)#
Router(config)#netwo
Router(config)#network 10.1.1.0 255.255.255.0
Router(config)#
^
% Invalid input detected at '^' marker.

Router(config)#ip dh
Router(config)#ip dhcp pool cisco
Router(dhcp-config)#network 10.1.1.0 255.255.255.0
Router(dhcp-config)#dns-
Router(dhcp-config)#dns-server 1.1.1.1
Router(dhcp-config)#
Router(dhcp-config)#def
Router(dhcp-config)#default-router 10.1.1.1
Router(dhcp-config)#
Router(dhcp-config)#
Router(dhcp-config)#
Router(dhcp-config)#|
```

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Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(dhcp-config)#
Router(dhcp-config)#sh ip
Router(dhcp-config)#exx
Router(dhcp-config)#
^
% Invalid input detected at '^' marker.

Router(dhcp-config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

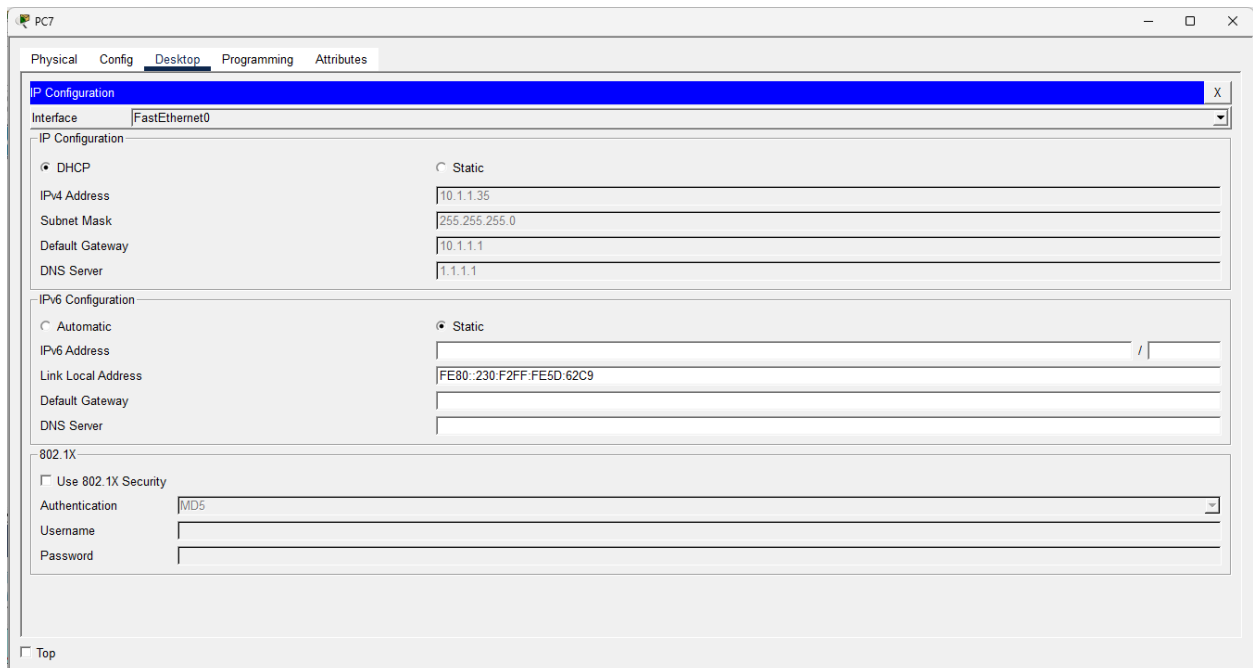
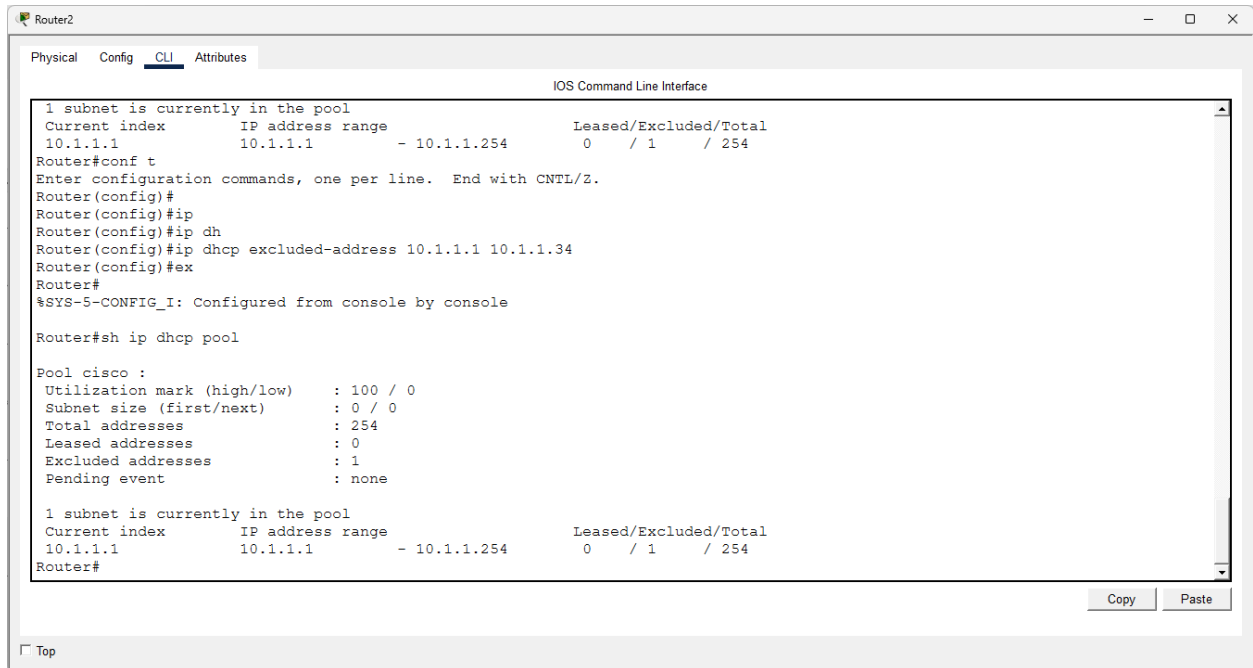
Router#
Router#sh ip dhcp po
Router#sh ip dhcp pool
Router#sh ip dhcp pool

Pool cisco :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)          : 0 / 0
Total addresses                   : 254
Leased addresses                  : 0
Excluded addresses                : 1
Pending event                     : none

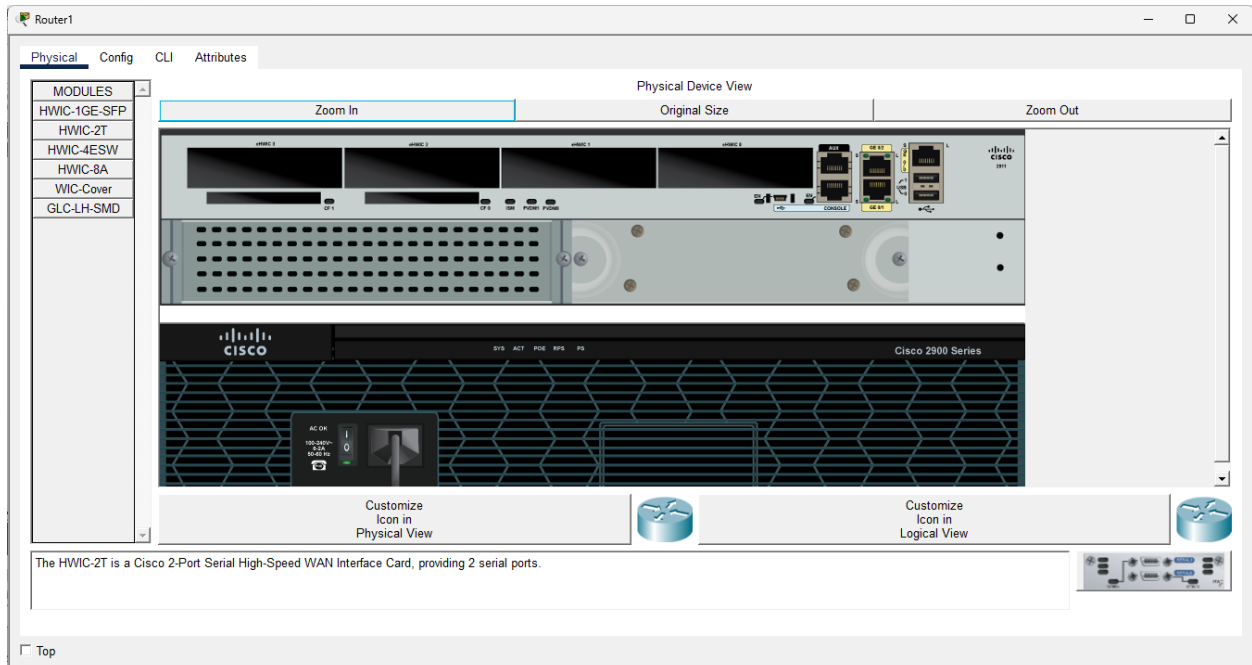
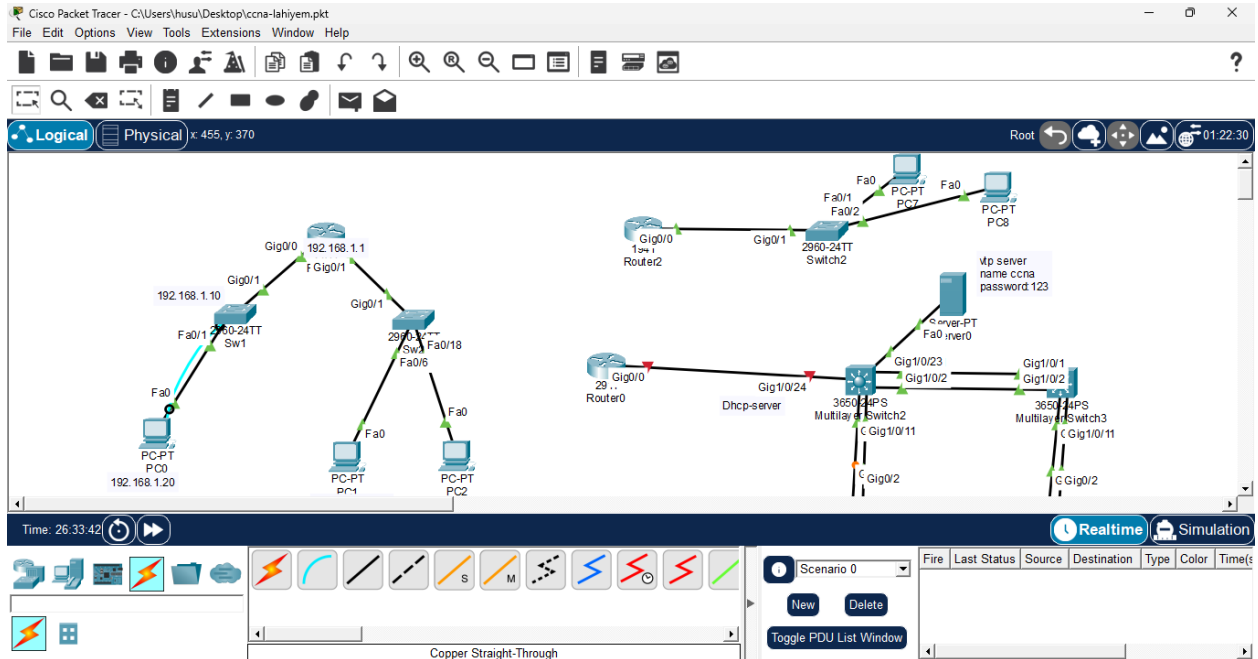
1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
10.1.1.1          10.1.1.1 - 10.1.1.254    0 / 1 / 254
Router#
```

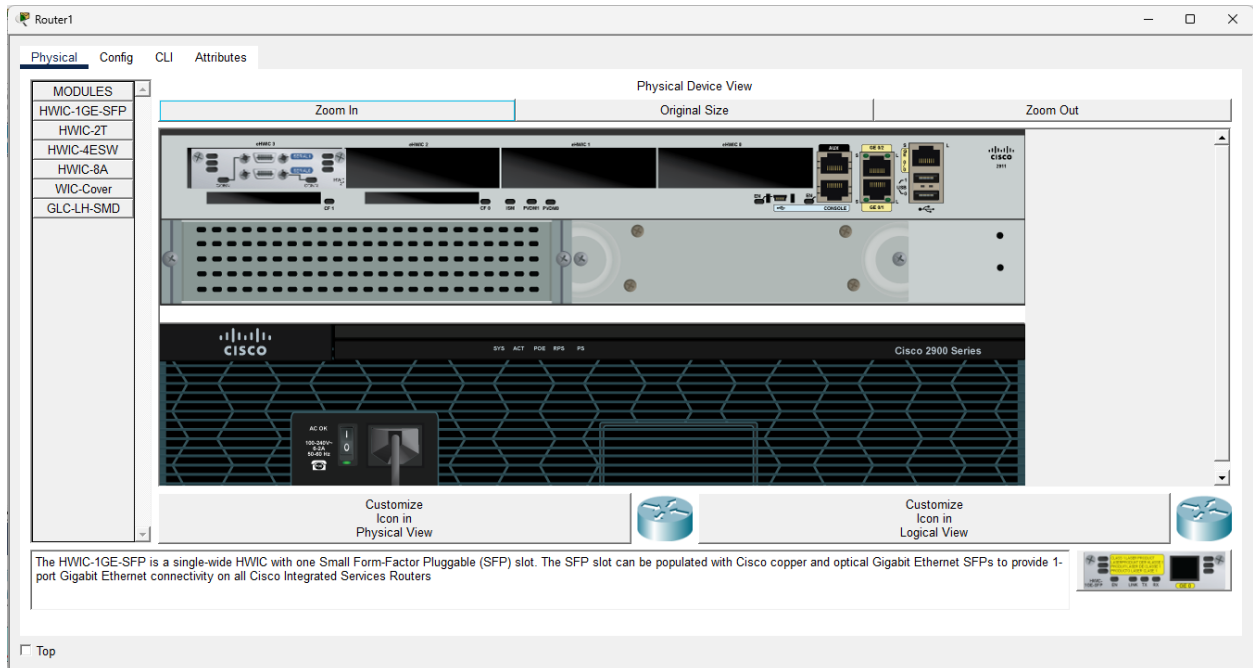
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Routing ucun arxitekturanin qurulmasi





Routerların serial interfeyslere ip lerin verilmesi



Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up
Icazesiz giris qadagandir

R1>
R1>
R1>en
Password:
Password:
R1#
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#
R1(config)#int ser
R1(config)#int ser
R1(config)#int serial 0/3/1
R1(config-if)#ip add
R1(config-if)#no sh
R1(config-if)#
R1(config-if)#
R1(config-if)#ip ad
R1(config-if)#ip address 209.165.203.22 255.255.255.0
R1(config-if)#
R1(config-if)#
R1(config-if)#ex
R1(config)#
```

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Multilayer Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/24, changed state to up

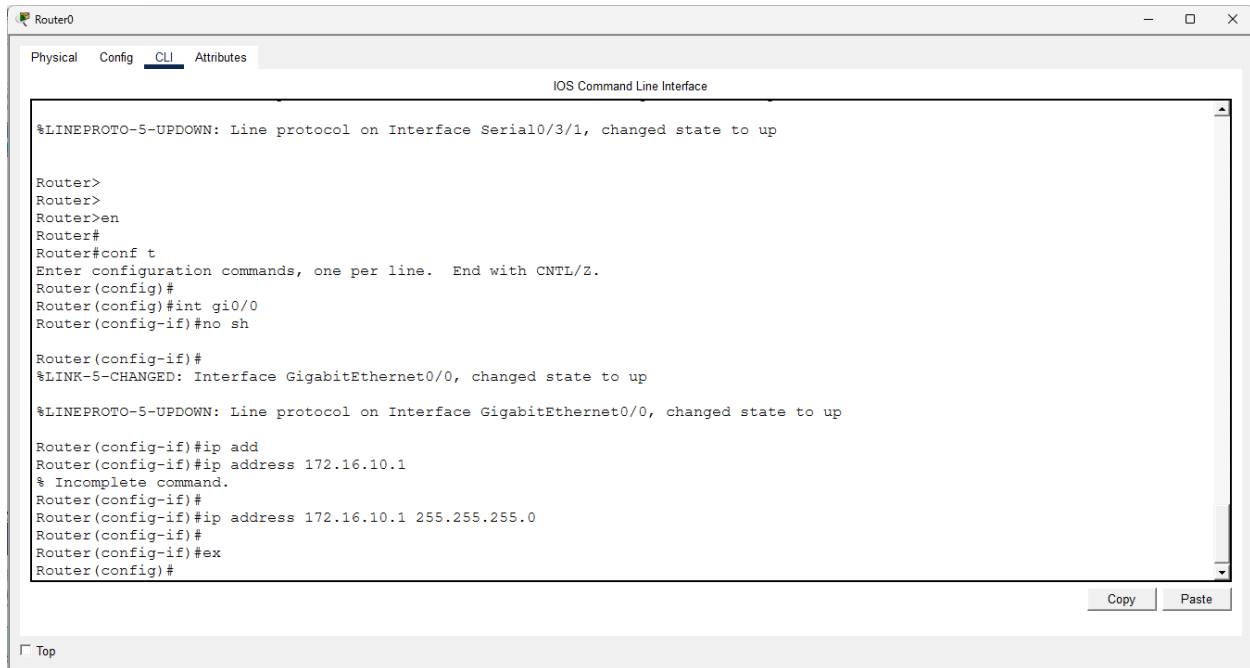
%LINK-3-UPDOWN: Interface GigabitEthernet1/0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/24, changed state to down

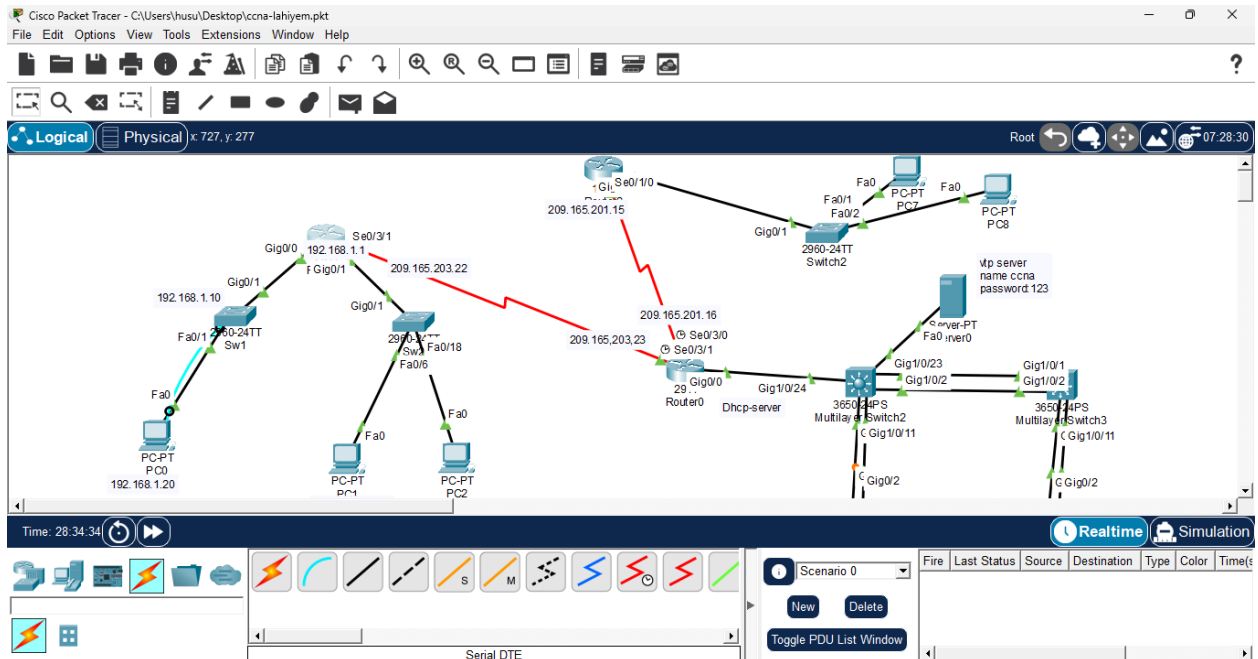
DS1>
DS1>
DS1>en
DS1#
DS1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
DS1(config)#
DS1(config)#int gi1/0/24
DS1(config-if)#
DS1(config-if)#ip ad
DS1(config-if)#ip add
DS1(config-if)#swi
DS1(config-if)#no switchport
DS1(config-if)#
DS1(config-if)#
DS1(config-if)#ip ad
DS1(config-if)#ip address 172.16.10.2 255.255.255.0
DS1(config-if)#
DS1(config-if)#ex
DS1(config)#
```

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Top



Eigrp ile routing



Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
tech-support      Show system information for Tech-Support
terminal          Display terminal configuration parameters
users             Display information about terminal lines
version           System hardware and software status
vlan-switch       VTP VLAN status
vtp               Configure VLAN database
zone              Zone Information
zone-pair         Zone pair information

R1#sh ip ro
R1#sh ip route ?
WORD              Network to display information about or hostname
bgp               Border Gateway Protocol (BGP)
connected         Connected
eigrp             Enhanced Interior Routing Protocol (EIGRP)
ospf              Open Shortest Path First (OSPF)
rip               Routing Information Protocol (RIP)
static            Static routes
summary           Summary of all routes
|                Output Modifiers
<cr>

R1#sh ip route co
C 192.168.1.0/24   is directly connected, GigabitEthernet0/0
C 192.168.100.0/24 is directly connected, GigabitEthernet0/1.100
C 192.168.200.0/24 is directly connected, GigabitEthernet0/1.200
C 209.165.203.0/24 is directly connected, Serial0/3/1

R1#
```

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Top

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
WORD              Network to display information about or hostname
bgp               Border Gateway Protocol (BGP)
connected         Connected
eigrp             Enhanced Interior Routing Protocol (EIGRP)
ospf              Open Shortest Path First (OSPF)
rip               Routing Information Protocol (RIP)
static            Static routes
summary           Summary of all routes
|                Output Modifiers
<cr>

R1#sh ip route co
C 192.168.1.0/24   is directly connected, GigabitEthernet0/0
C 192.168.100.0/24 is directly connected, GigabitEthernet0/1.100
C 192.168.200.0/24 is directly connected, GigabitEthernet0/1.200
C 209.165.203.0/24 is directly connected, Serial0/3/1

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#rou
R1(config)#router ei
R1(config)#router eigrp 100
R1(config-router)#net 192.168.1.0
R1(config-router)#net 192.168.100.0
R1(config-router)#net 192.168.200.0
R1(config-router)#net 209.165.203.0
R1(config-router)#
R1(config-router)#
```

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Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config)#r0
Router(config)#router ei
Router(config)#router eigrp 100
Router(config-router)#net 172.16.10.0
Router(config-router)#ex
Router(config)#int ser 0.3.1
^
% Invalid input detected at '^' marker.

Router(config)#int ser
Router(config)#int serial 0/3/1
Router(config-if)#ip add
Router(config-if)#ip address 209.165.203.23 255.255.255.0
Router(config-if)#
Router(config-if)#ex
Router(config)#do sh ip rou conn
C 172.16.10.0/24 is directly connected, GigabitEthernet0/0
C 209.165.203.0/24 is directly connected, Serial0/3/1

Router(config)#router eigrp 100
Router(config-router)#net 209.165.203.0
Router(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 209.165.203.22 (Serial0/3/1) is up: new adjacency

Router(config-router)#
Router(config-router)#
Router(config-router)#
```

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Router2

Physical Config CLI Attributes

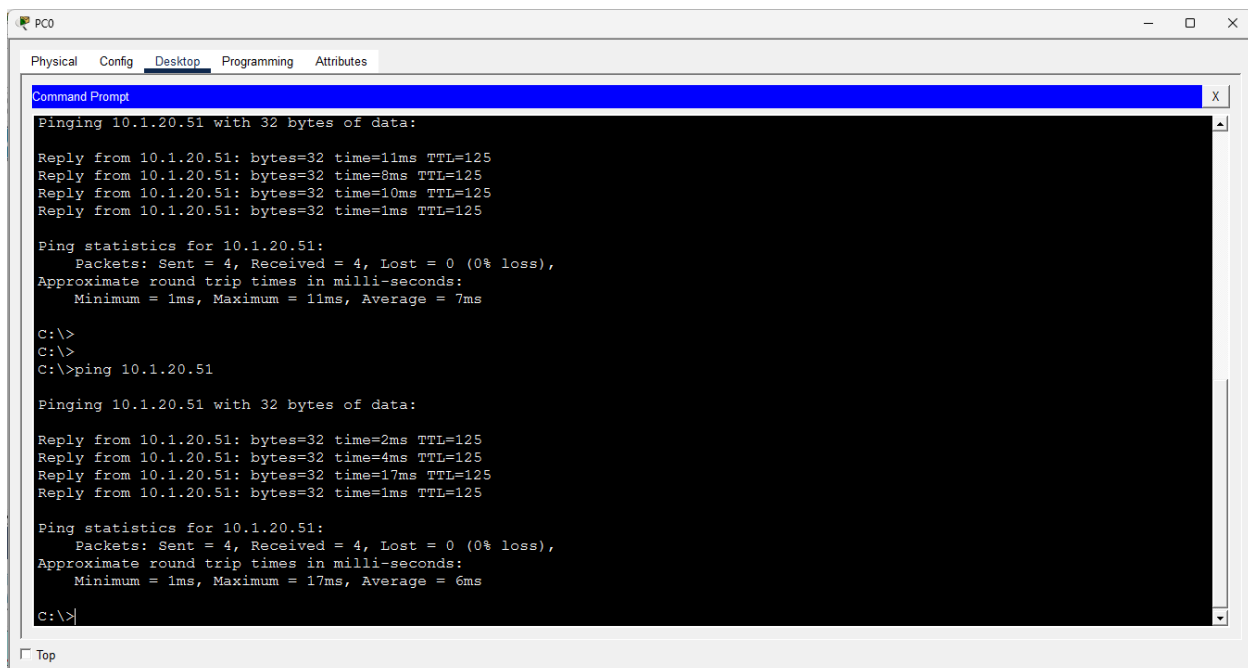
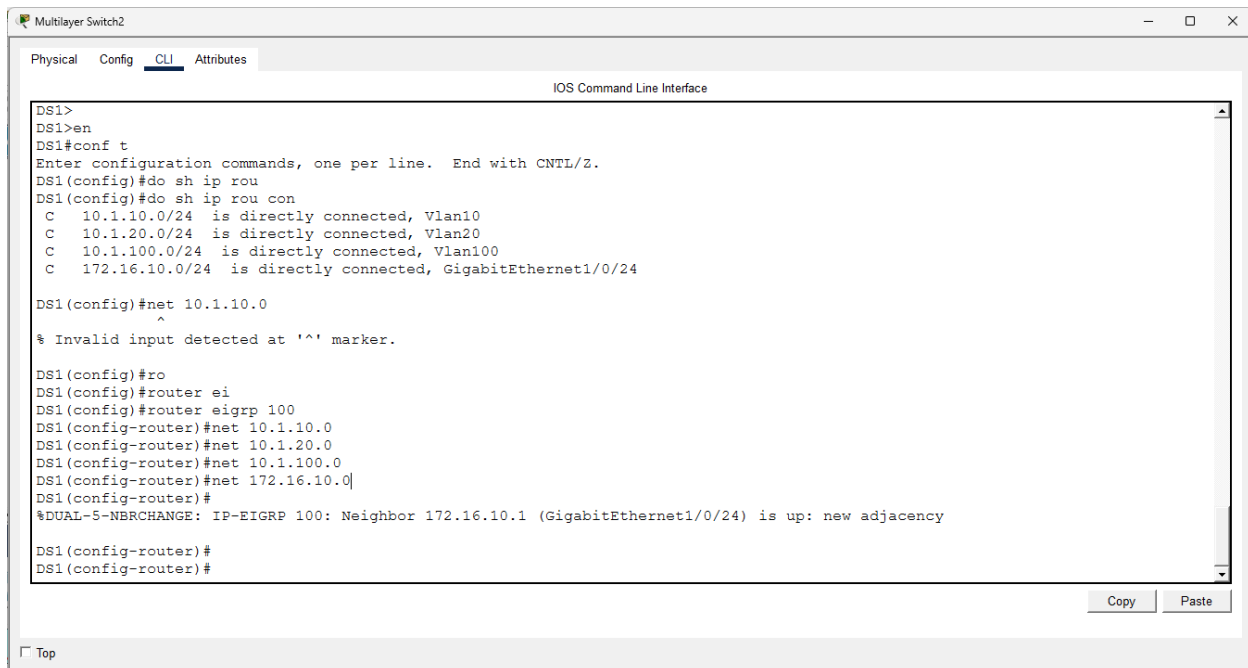
IOS Command Line Interface

```
Router>
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#do sh ip rou con
C 10.1.1.0/24 is directly connected, GigabitEthernet0/0
C 209.165.201.0/24 is directly connected, Serial0/1/0

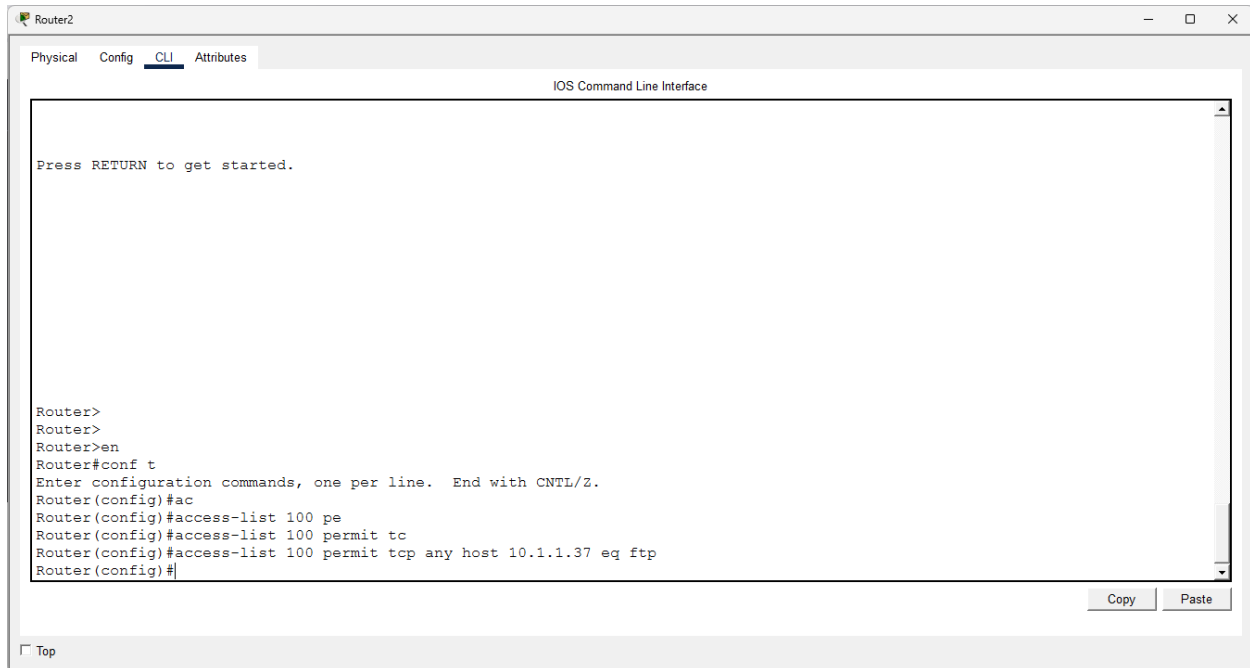
Router(config)#rou
Router(config)#router ei
Router(config)#router eigrp 100
Router(config-router)#net 10.1.1.0
Router(config-router)#net 209.165.201.0
Router(config-router)#ex
Router(config)#
Router(config)#
```

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Acl konfigurasiyasi



Router2

Physical Config CLI Attributes

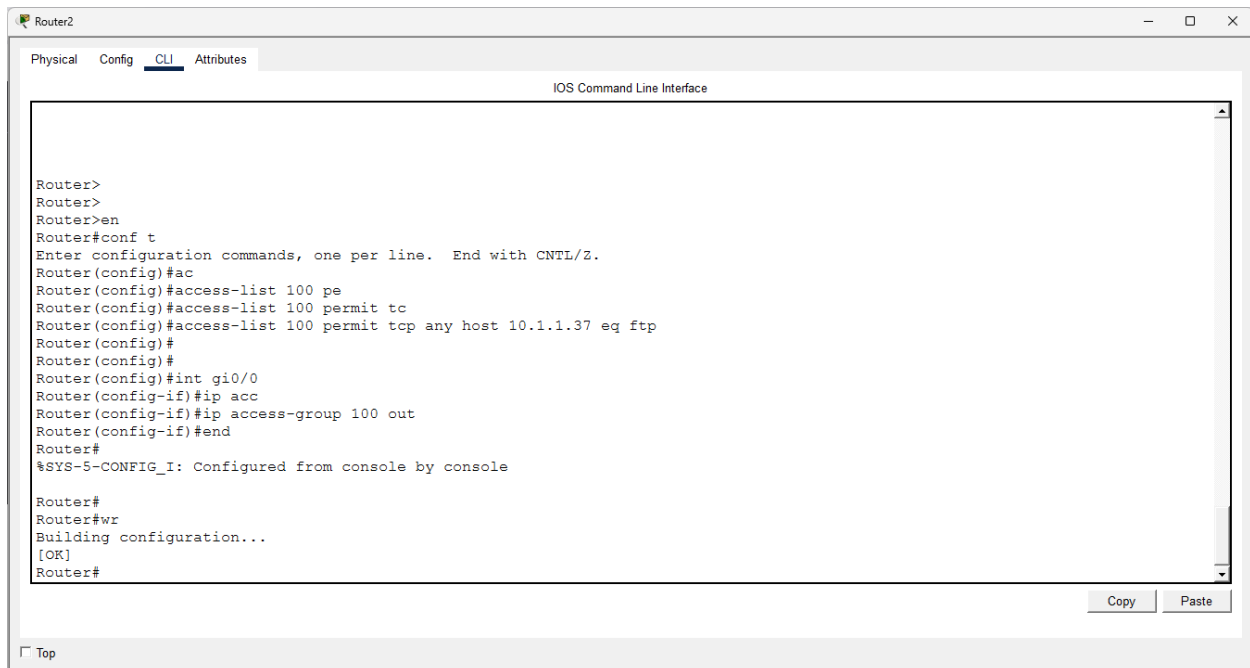
IOS Command Line Interface

Press RETURN to get started.

```
Router>
Router>
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ac
Router(config)#access-list 100 pe
Router(config)#access-list 100 permit tc
Router(config)#access-list 100 permit tcp any host 10.1.1.37 eq ftp
Router(config)#
```

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Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>
Router>
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ac
Router(config)#access-list 100 pe
Router(config)#access-list 100 permit tc
Router(config)#access-list 100 permit tcp any host 10.1.1.37 eq ftp
Router(config)#
Router(config)#
Router(config)#int gi0/0
Router(config-if)#ip acc
Router(config-if)#ip access-group 100 out
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#wr
Building configuration...
[OK]
Router#
```

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Nəticə

Bu laboratoriya layihəsi çərçivəsində Cisco CCNA proqramında öyrənilən əsas şəbəkə texnologiyaları praktiki şəkildə tətbiq edilmişdir. Router və switch cihazlarının ilkin konfigurasiyası həyata keçirilmiş, VLAN-lar yaradılmış, Inter-VLAN Routing, EtherChannel, Trunk, VTP, DHCP, EIGRP, STP, Port Security, SSH və ACL texnologiyaları uğurla tətbiq olunmuşdur.

Layihənin icrası zamanı şəbəkənin təhlükəsizliyi, etibarlılığı və idarə olunması istiqamətində praktiki təcrübə qazanılmış, yaranan problemlərin diaqnostikası və aradan qaldırılması üzrə bacarıqlar inkişaf etdirilmişdir.

Bu layihə Cisco IOS mühitində müəssisə şəbəkələrinin qurulması və idarə olunması üzrə əldə olunan praktik biliklərin nümayişi məqsədini daşıyır.

Qazanılan bacarıqlar

- ✓ Cisco IOS ilə işləmək
- ✓ Router və Switch konfigurasiyası
- ✓ VLAN və Inter-VLAN Routing
- ✓ EtherChannel
- ✓ Trunk

- ✓ VTP
- ✓ DHCP
- ✓ EIGRP
- ✓ STP
- ✓ Port Security
- ✓ SSH təhlükəsizliyi
- ✓ ACL
- ✓ Şəbəkə problemlərinin diaqnostikası və aradan qaldırılması